notebook

August 26, 2025

1 To do

- Find the alignment problem of 150 ms (motor?) in group 4 (experiment=2AFC_4)

2 Load behavioral data

3 Filter data

```
211k trials
```

4 Add lick data

```
0.16% of premature lick trials (before response window)
Port1In or Port2In are already lists
['absILD', 'LicksLeft', 'LicksRight', 'nLicksLeft', 'nLicksRight', 'nLicks',
'leftILI', 'rightILI', 'ILI', 'RT', 'RT2', 'SessionHalf']
```

	Trial	Side	RepTrial	Reward	Punish	Miss	WrongLick	Hit	AfterHit	\
0	0	0	NaN	1	0	0	NaN	1.0	NaN	
1	1	0	1.0	1	0	0	NaN	1.0	1.0	
2	2	1	0.0	0	1	0	NaN	0.0	1.0	
3	3	1	0.0	0	1	0	NaN	0.0	0.0	
4	4	1	0.0	1	0	0	NaN	1.0	0.0	

	Choice	…	LicksRight	nLicksLeft	\
0	0.0	…	[]	10	
1	0.0	…	[]	7	
2	0.0	…	[1.4690999999999996, 1.5776]	6	
3	0.0	…	[1.3843999999999999, 1.3899000000000004, 1.501…	2	
4	1.0	…	[1.3199, 1.4406000000000003, 1.5311999999999999…	0	

	nLicksRight	nLicks	leftILI	rightILI	ILI	RT	RT2	\
0	0	10	0.110156	NaN	0.110156	0.1525	0.1525	
1	0	7	0.110783	NaN	0.110783	0.0178	0.0178	
2	2	8	0.176000	0.10850	0.142250	0.1735	0.1735	
3	7	9	0.571800	0.11730	0.344550	0.0997	0.0997	
4	9	9	NaN	0.11755	0.117550	0.1521	0.1521	

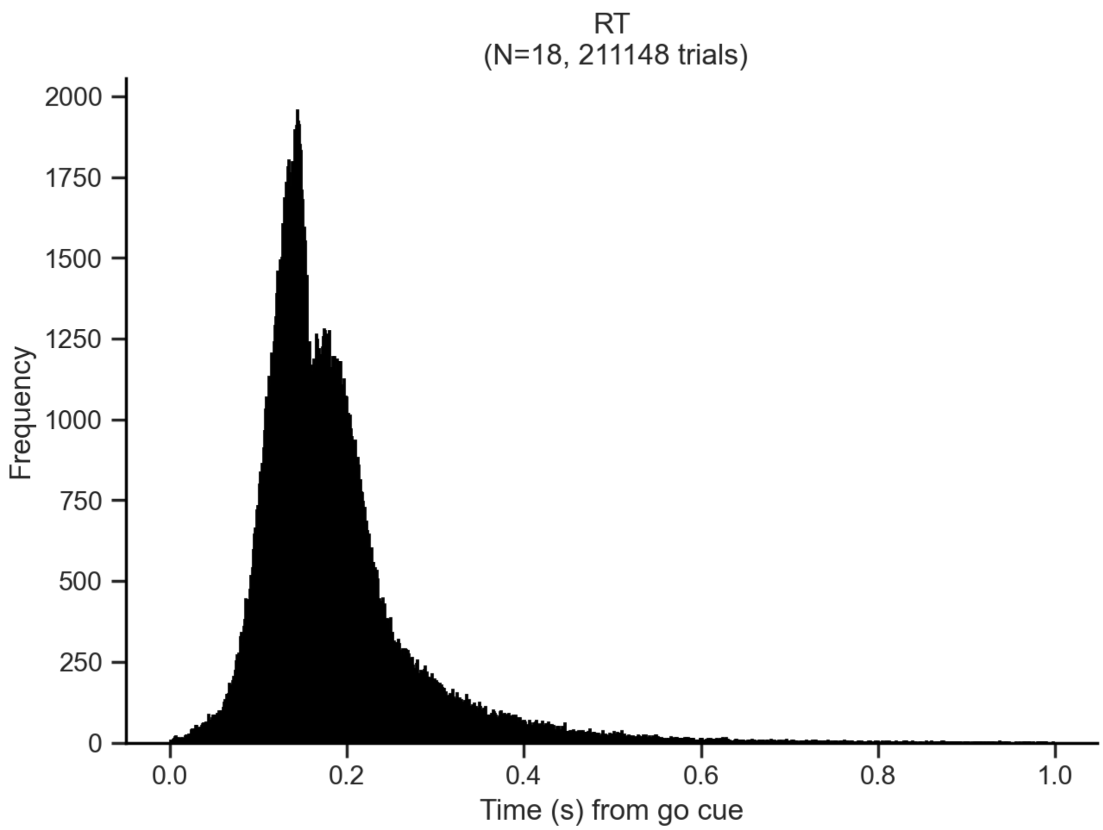
```
   SessionHalf
0            0
1            0
2            0
3            0
4            0

[5 rows x 86 columns]
```

5 Reaction Times (RTs)

5.1 Distributions

5.1.1 All trials

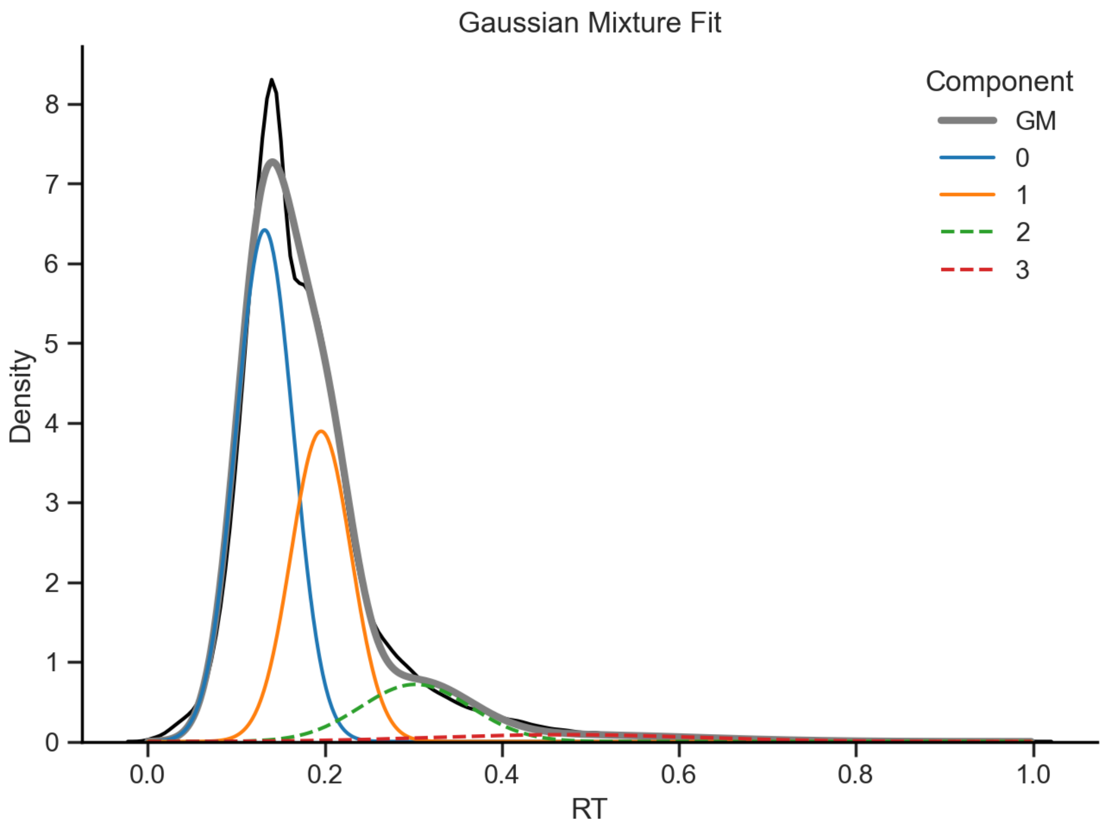


This is the rawest representation of RT data. There is a bimodal distribution of RTs across all mice. The overall distribution has a shoulder-like second peak (lower) after the first one. It is likely due to the presence of two different strategies in the task: one that involves a fast response and another that involves a slower response. Let's rule out other hypotheses first (easier and faster to check).

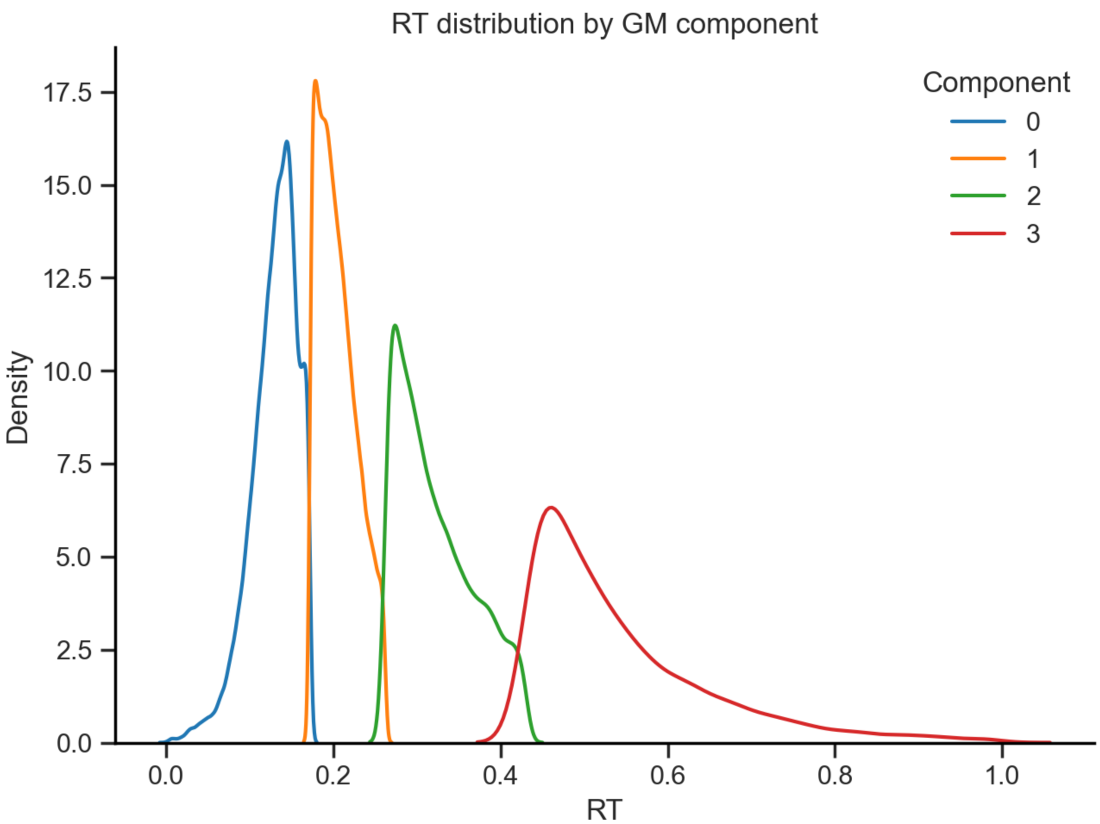
One hypothesis is that the difference evidence levels (ILDs) drives different RTs. To examine this, we can plot a RT distribution per ILD level. To visually compare the distributions, a Kernel Desnity Estimation (KDE) plot is preferred.

Gaussian Mixture fit

```
Component 0: mean=0.13 s, std=0.03 s, weight=0.52
Component 1: mean=0.20 s, std=0.03 s, weight=0.33
Component 2: mean=0.30 s, std=0.06 s, weight=0.11
Component 3: mean=0.47 s, std=0.15 s, weight=0.03
```



Plot RT distributions grouped by component



Screening of variables to explain the difference between the first 2 components (0 and 1)

```
Index(['Trial', 'Side', 'RepTrial', 'Reward', 'Punish', 'Miss', 'WrongLick',
       'Hit', 'AfterHit', 'Choice', 'RepChoice', 'Response', 'TrialStart',
       'TrialEnd', 'TrialLen', 'StimStart', 'StimEnd', 'StimLen',
       'RespWinStart', 'RespWinEnd', 'RespWinLen', 'Filename', 'Filename2',
       'FilesMatch', 'Message', 'MessageFound', 'SoundLeft', 'SoundRight',
       'Sound', 'ILD', 'absILD', 'ILDRep', 'Port1In', 'Port1Out', 'Port2In',
       'Port2Out', 'ValveLeft', 'ValveRight', 'AW', 'AWTrials', 'Switch',
       'Timeout', 'Fixation', 'Stage', 'Motor', 'REC', 'Progression', 'CB',
       'RespWin', 'ITI', 'WarmUp', 'RecoveryMode', 'P', 'PRight', 'Blocks',
       'BlockLen', 'Task', 'Ephys', 'StimDur', 'Delay', 'VarDelay',
       'SerialPort', 'Protocol', 'Creator', 'Project', 'Experiment', 'Board',
       'Setup', 'NetPort', 'Subject', 'BpodApiVersion', 'Session', 'Date',
       'SessionStart', 'SessionEnd', 'LicksLeft', 'LicksRight', 'nLicksLeft',
       'nLicksRight', 'nLicks', 'leftILI', 'rightILI', 'ILI', 'RT', 'RT2',
       'SessionHalf', 'RTlabelsGM'],
      dtype='object')
```

```
 Variable      Type   MeanDiff            pval       pval_adj
```

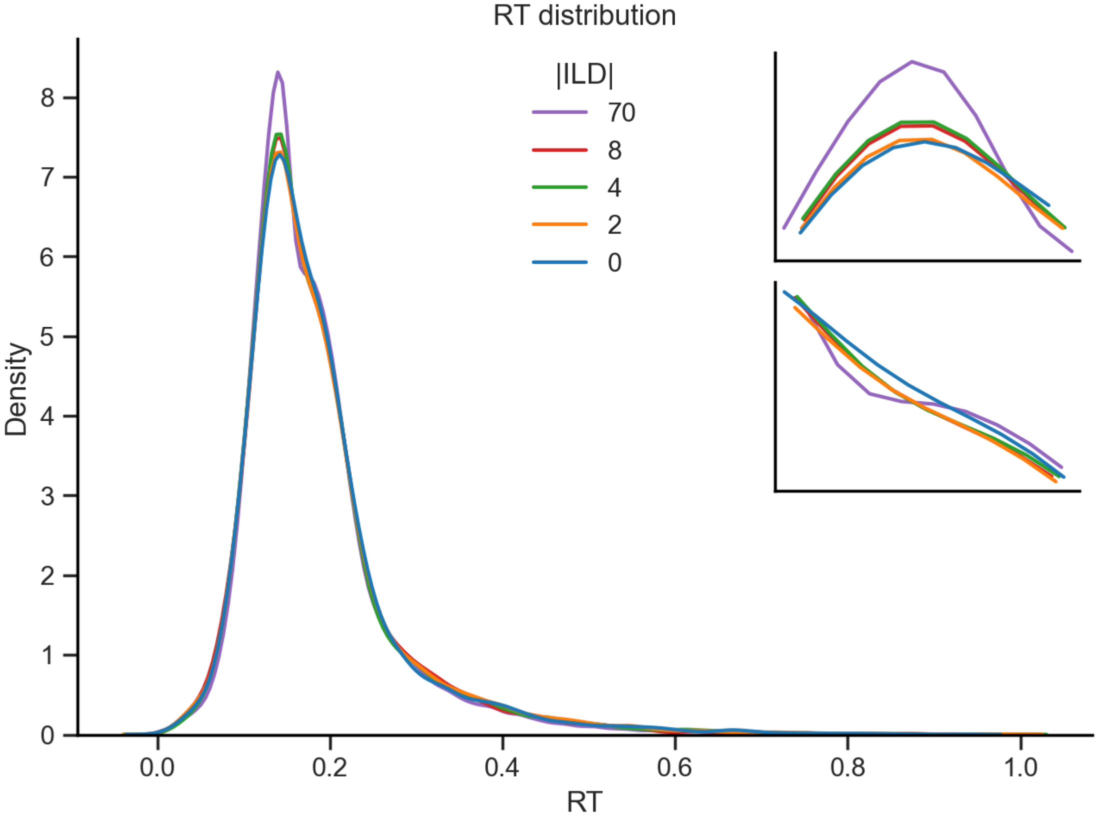
13	SessionHalf	numeric	-0.062288	2.405513e-148	3.367718e-147
11	nLicksRight	numeric	0.397858	5.741535e-110	8.038149e-109
10	nLicksLeft	numeric	0.295334	1.459094e-70	2.042731e-69
4	AfterHit	numeric	0.023858	2.896563e-37	4.055189e-36
3	Hit	numeric	0.022993	2.907457e-34	4.070440e-33
0	Trial	numeric	-8.350222	5.112720e-31	7.157808e-30
9	P	numeric	0.008306	4.070178e-25	5.698249e-24
7	ILD	numeric	2.432461	1.044159e-16	1.461823e-15
5	Choice	numeric	0.018747	5.819041e-15	8.146657e-14
1	Side	numeric	0.016705	3.596736e-12	5.035430e-11
8	ILDRep	numeric	1.841166	4.194612e-10	5.872457e-09
2	RepTrial	numeric	0.014033	6.436676e-09	9.011346e-08
6	RepChoice	numeric	0.008209	5.571591e-04	7.800227e-03
12	absILD	numeric	0.096950	4.760186e-01	6.664260e+00

5.1.2 Splits

ILD

Together

```
ILD 70: peak RT = 0.139 s
ILD 8: peak RT = 0.143 s
ILD 4: peak RT = 0.143 s
ILD 2: peak RT = 0.143 s
ILD 0: peak RT = 0.141 s
Mean RT = 0.142 s
```

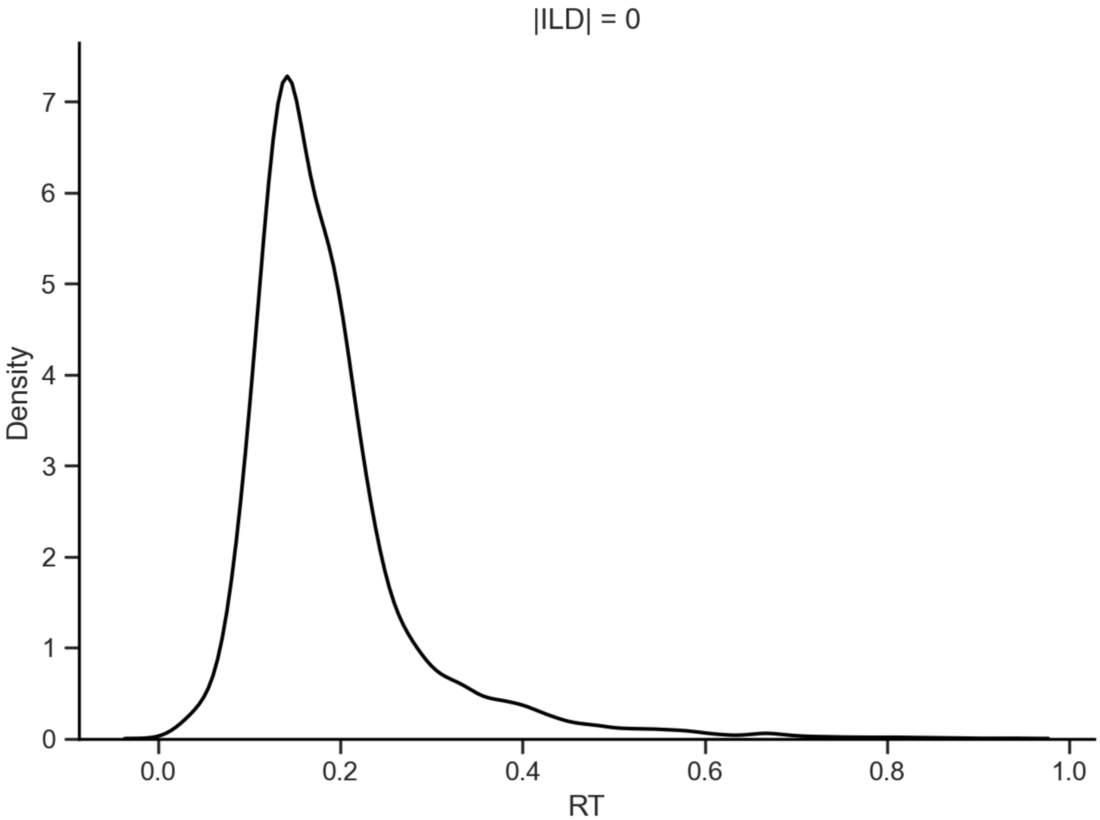


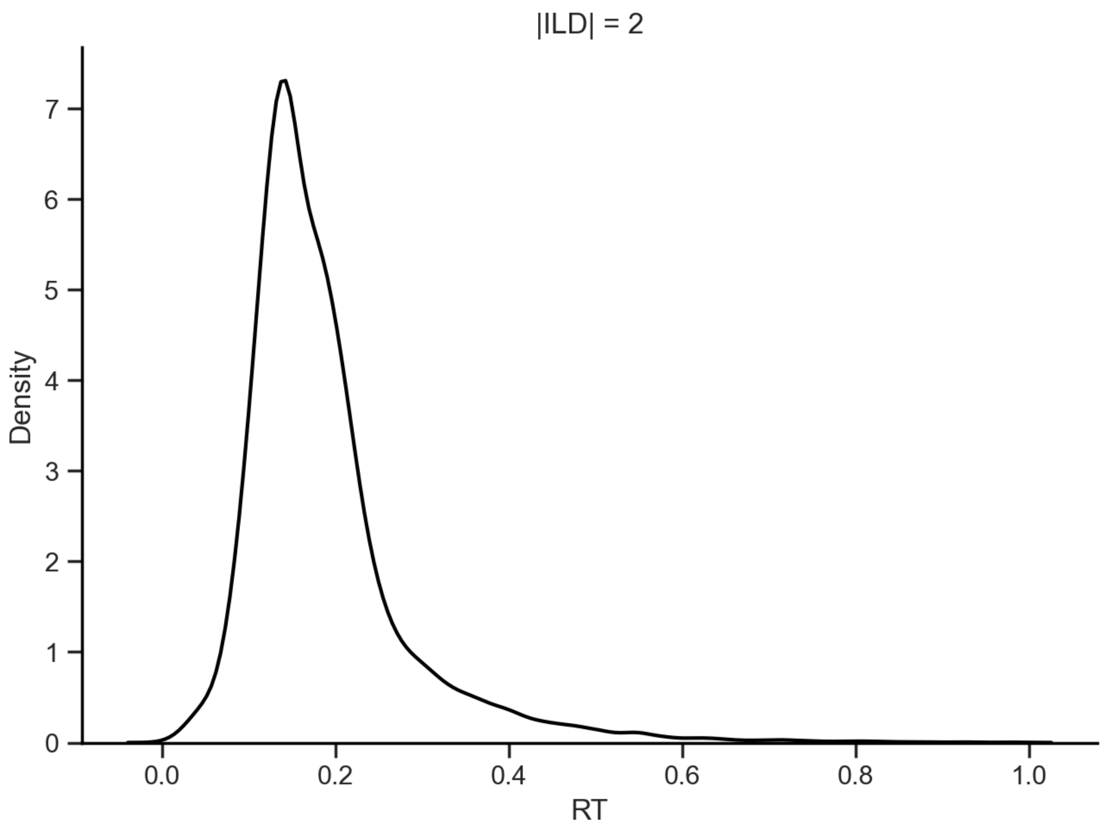
The ILD does not seem to explain the existence bimodality in the RT distribution, as the bimodality is present across all ILD levels.

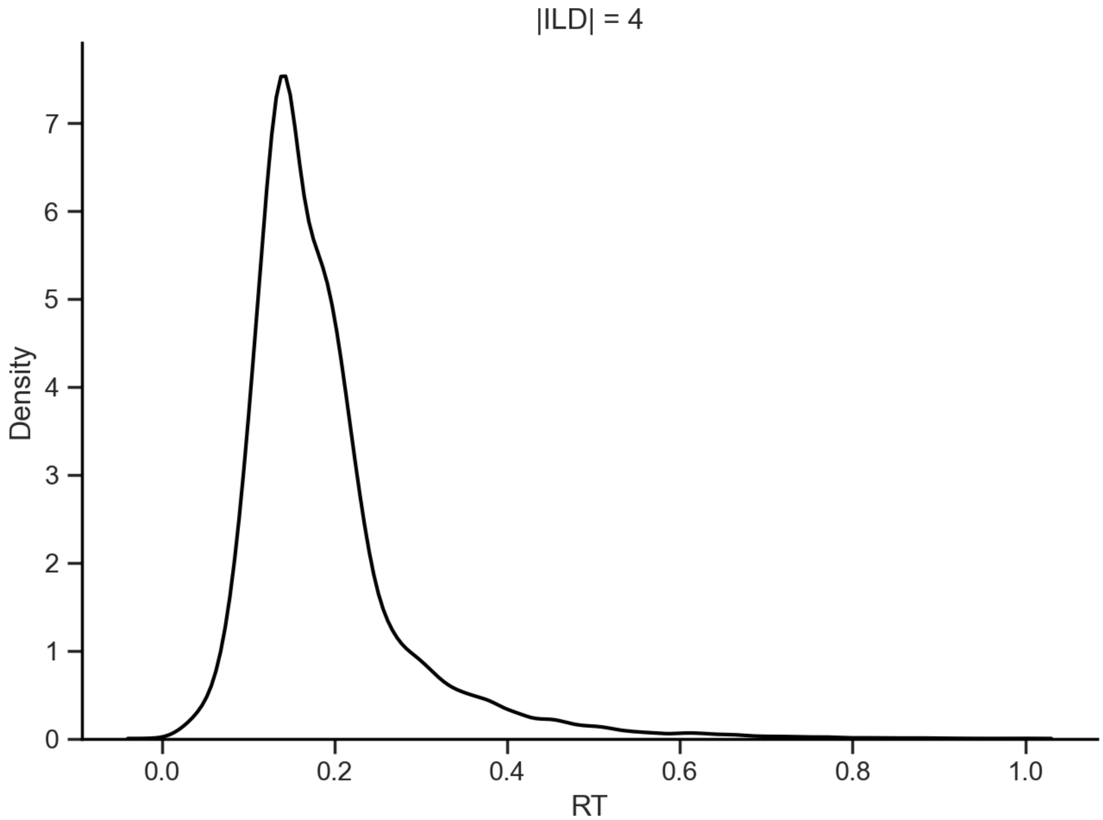
Nevertheless, zooming in the peak of the distribution (0.1-0.2 s), we can see that the RTs are faster for lower ILD levels, which is expected as the evidence is more certain. What is surprising is this modulation to show up despite the fact that mice were head-fixed. In addition, zoomin in the second peak of the distribution (0.15-0.2 s), we can see that the second peak is more pronounced for lower ILD levels, suggesting that the bimodality is actually modulated by ILD level.

Too many lines can make on top of each other make it hard to disentangle the ILD impact on RT, so we can plot the RT distribution for each ILD level separately.

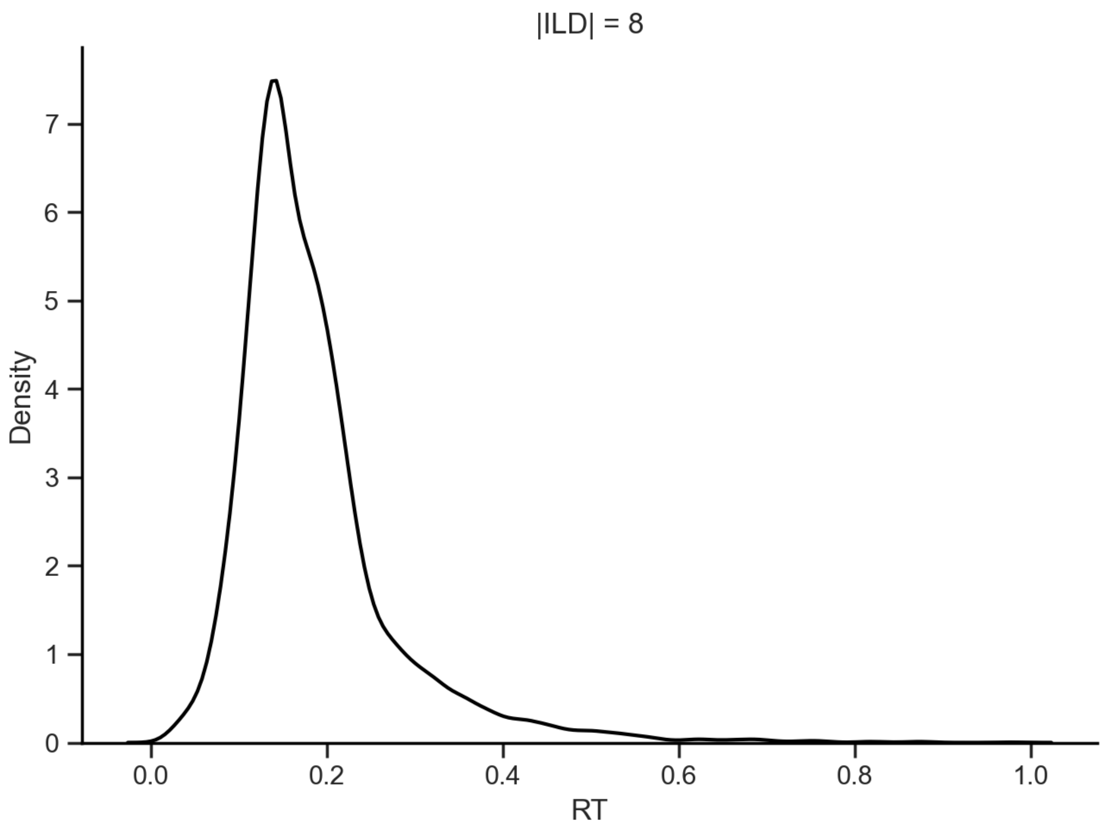
Individually

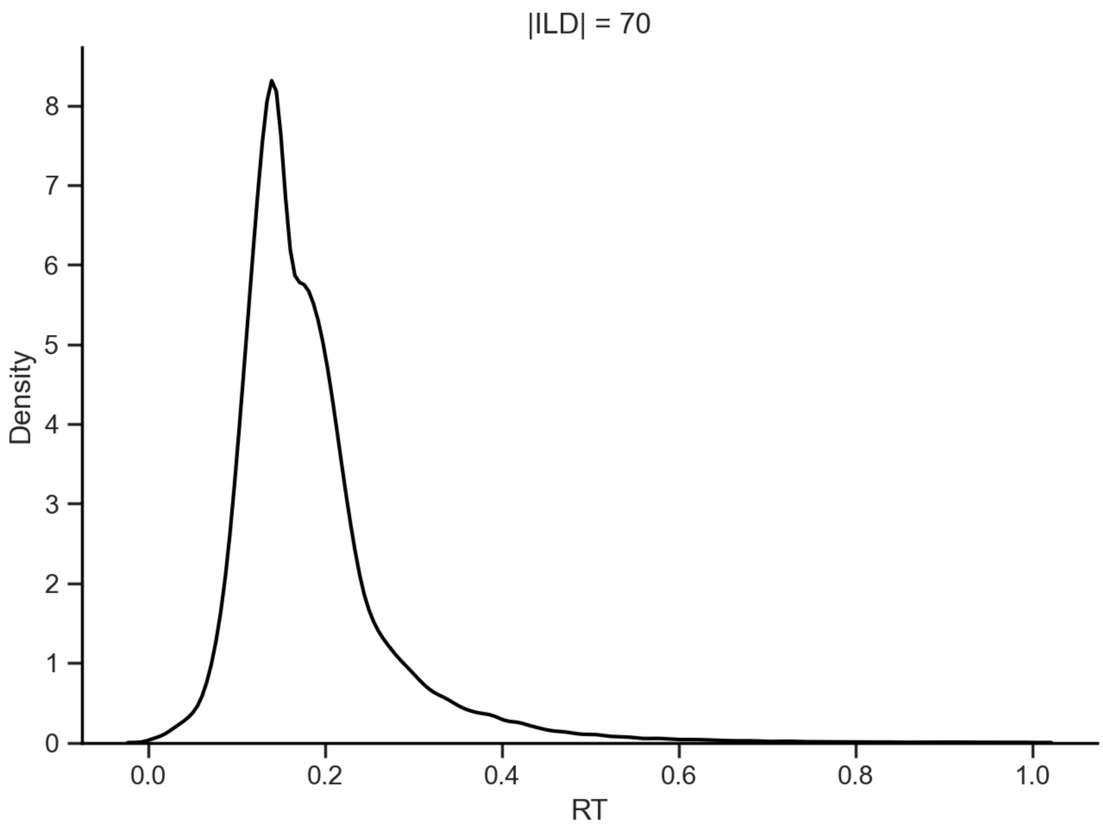

|ILD| = 0
Density
7
6
5
4
3
2
1
0
0.0
0.2
0.4
0.6
0.8
1.0
RT


|ILD| = 2
Density
7
6
5
4
3
2
1
0
0.0
0.2
0.4
0.6
0.8
1.0
RT



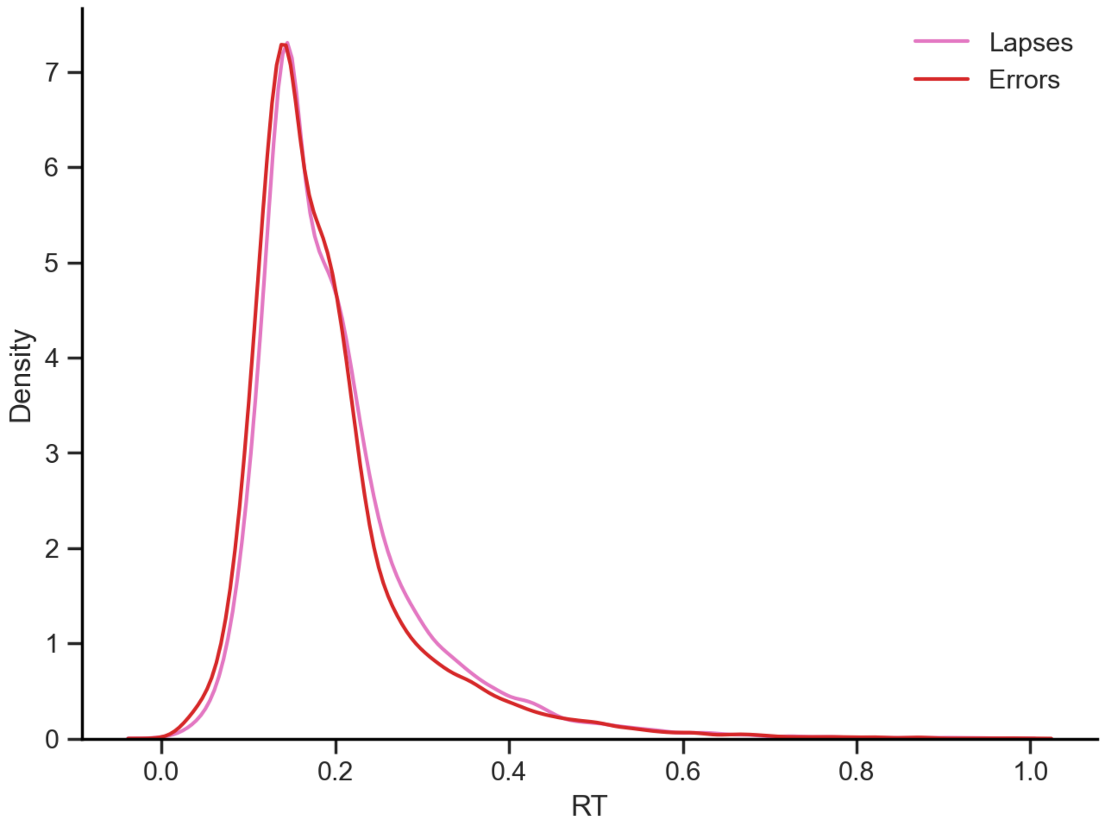
|ILD| = 4
Density
RT
0
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8
1.0

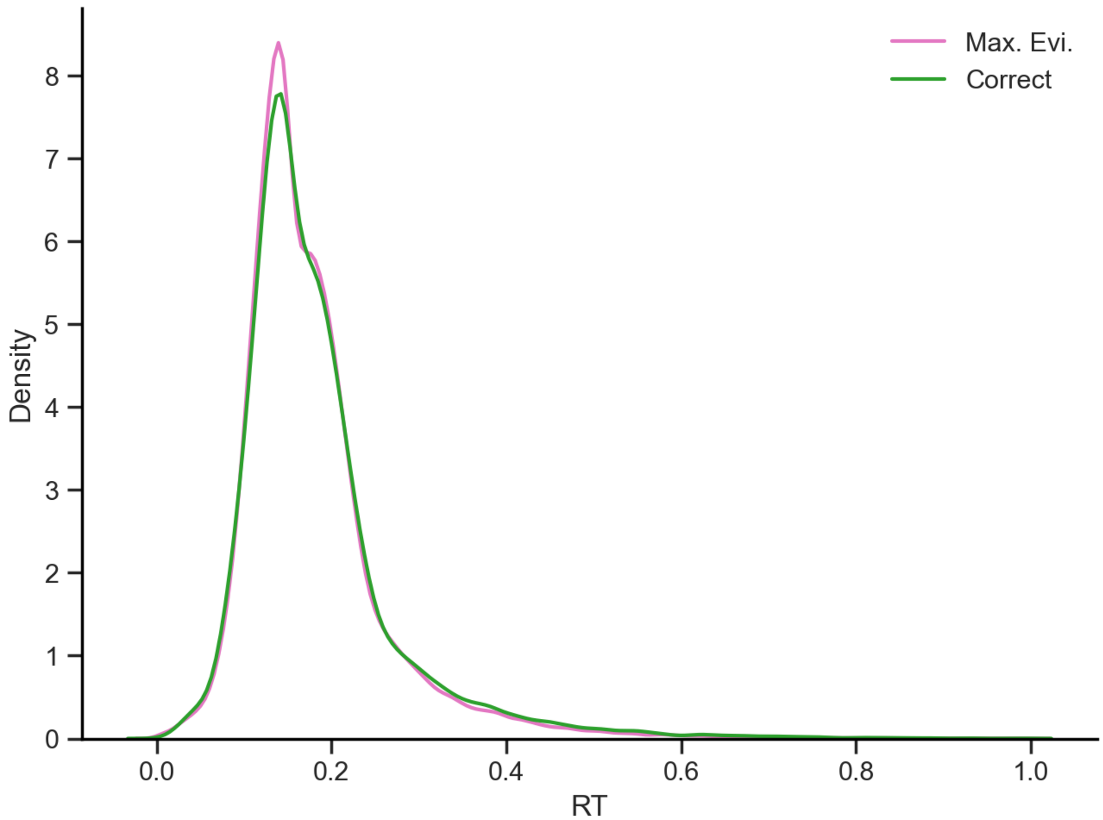

|ILD| = 8
Density
RT
0
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8
1.0



This is interesting. Although the bimodality is present in all ILD levels, the second peak is more pronounced for lower ILD levels. This suggests that the bimodality is not driven by differences in evidence levels, but rather by a different strategy used by the mice.

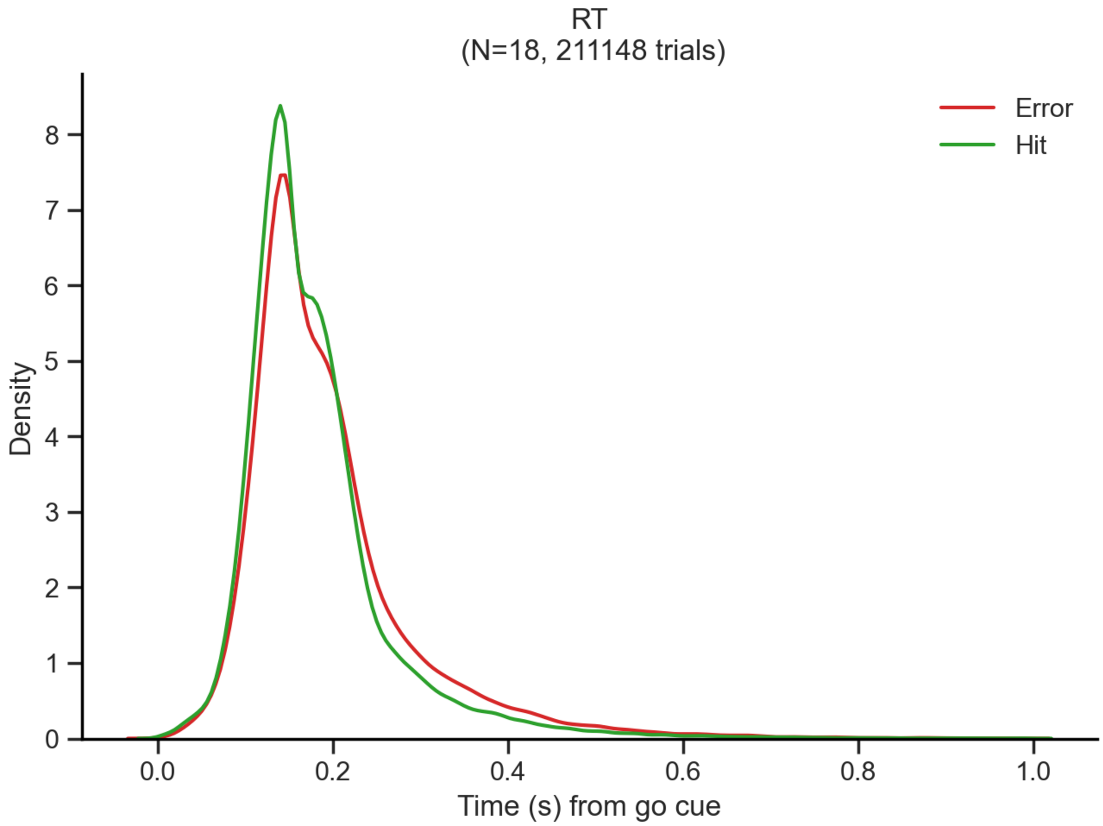
```
<matplotlib.legend.Legend at 0x15d0a955900>
```


Lapses
Errors
Density
RT
0
1
2
3
4
5
6
7
0.0
0.2
0.4
0.6
0.8
1.0

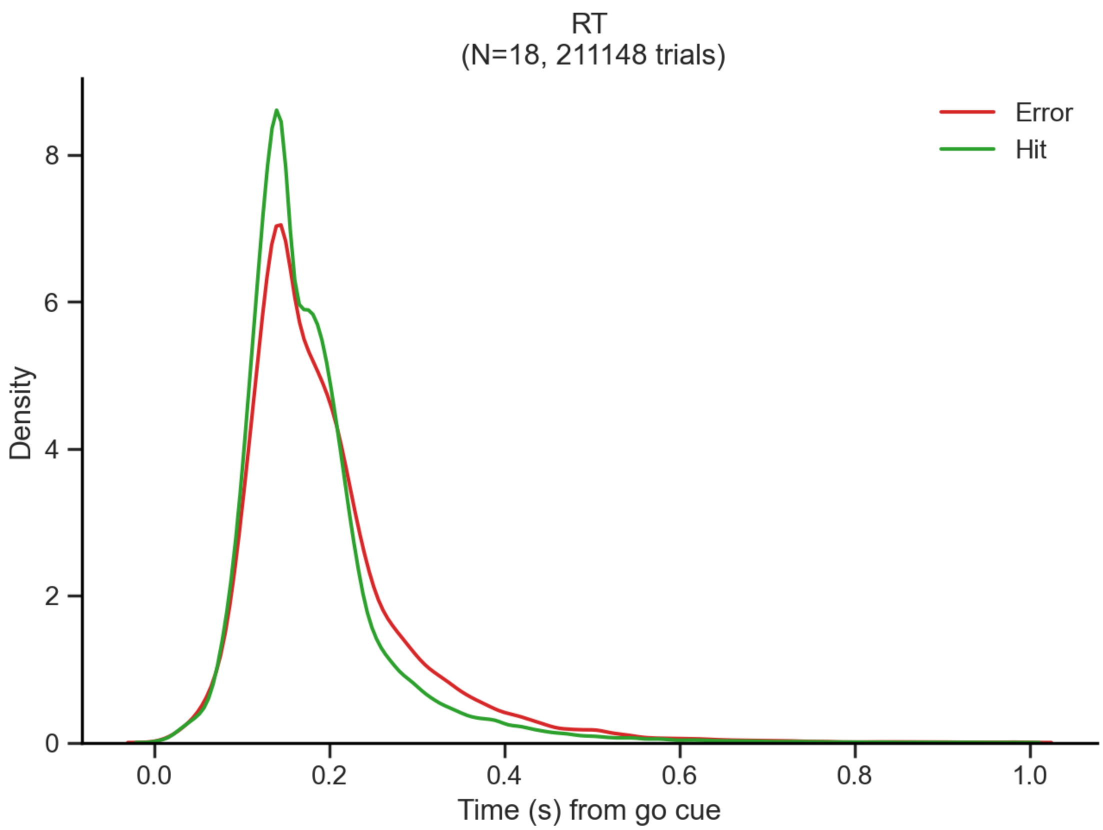


Another hypothesis is that the bimodality is driven by differences in behavioral variables such as outcome, choice, stimulus, repeat, or previous outcome. To examine this, we can plot a RT distribution per behavioral variable split in 2 conditions.

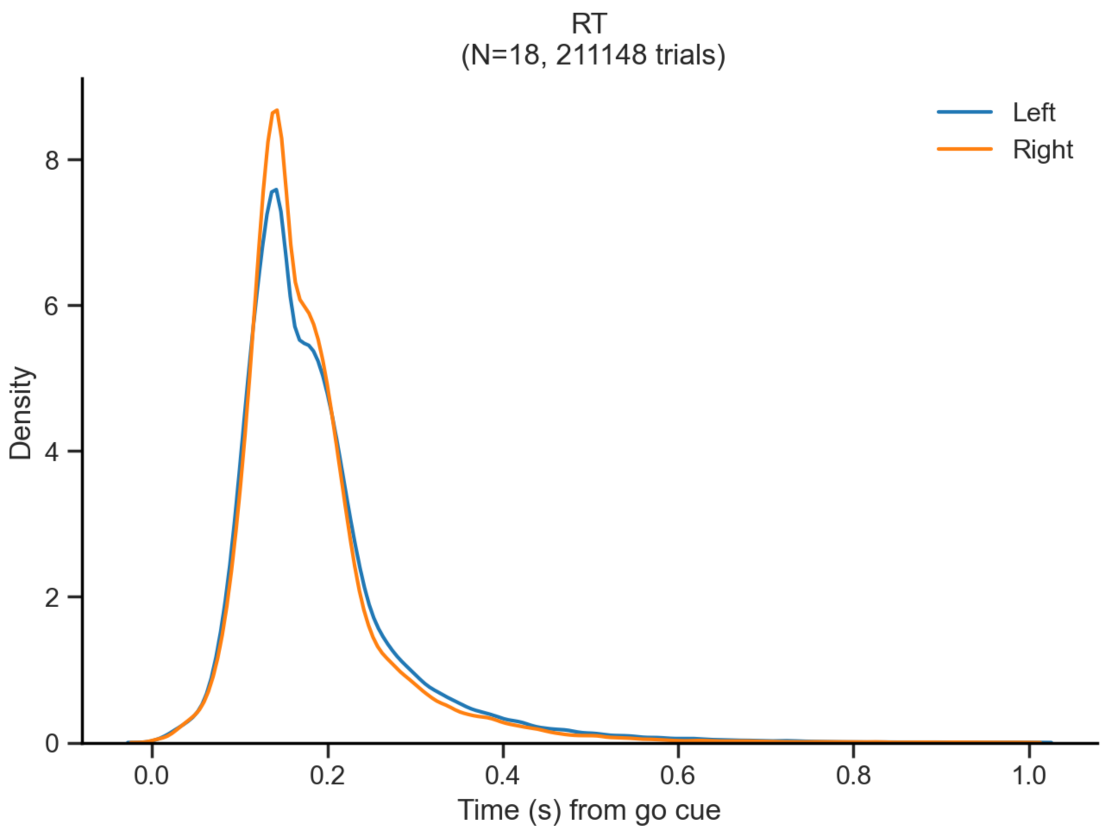
Outcome



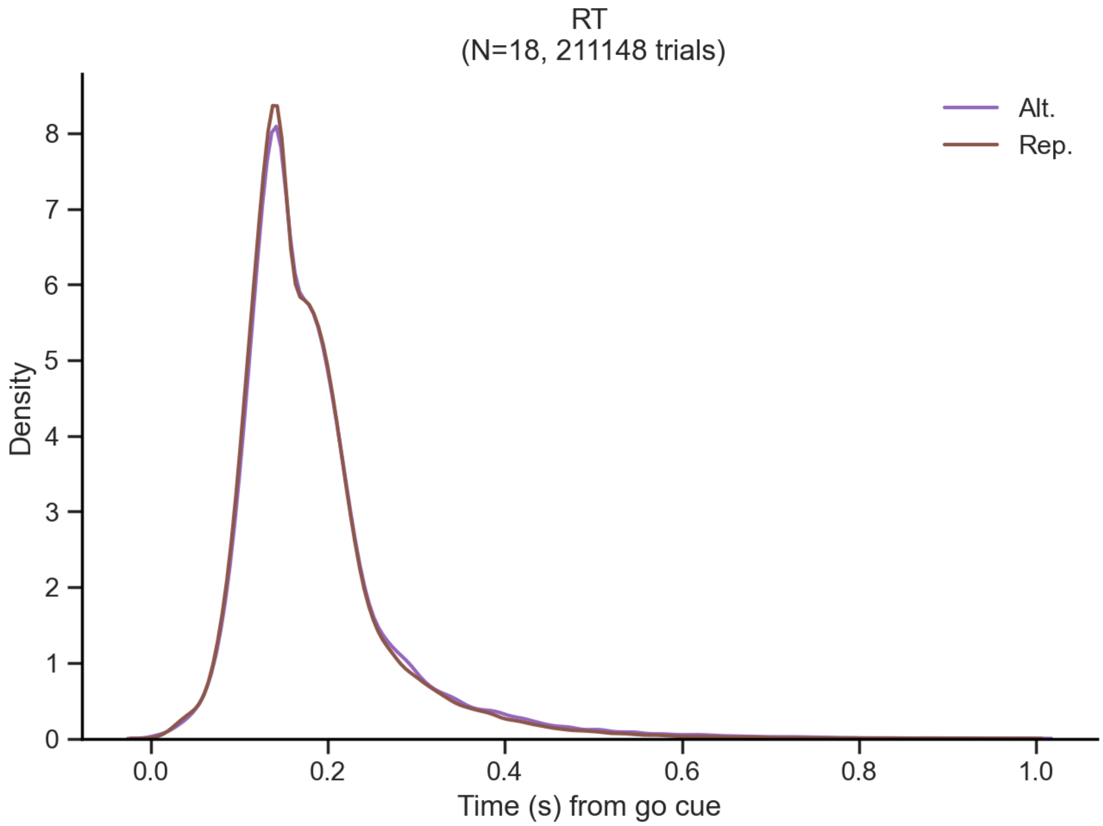
Previous outcome


RT
(N=18, 211148 trials)
Error
Hit
Density
8
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
Time (s) from go cue

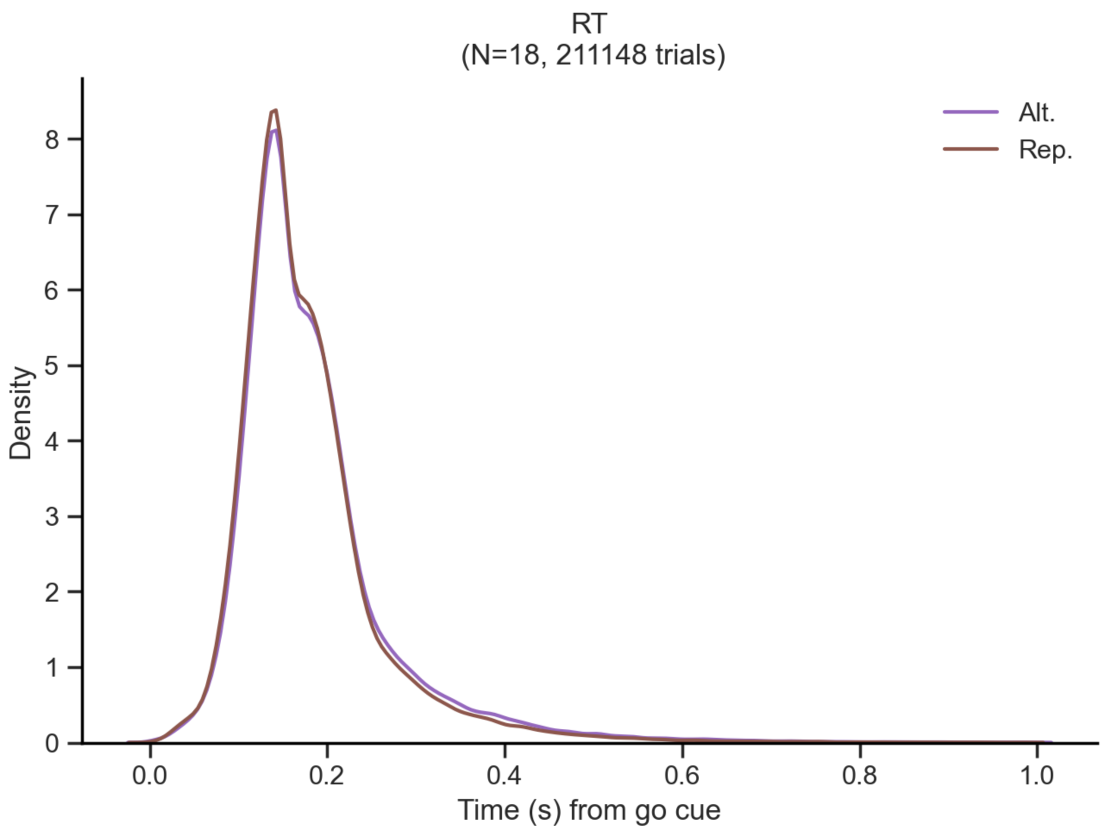
Choice



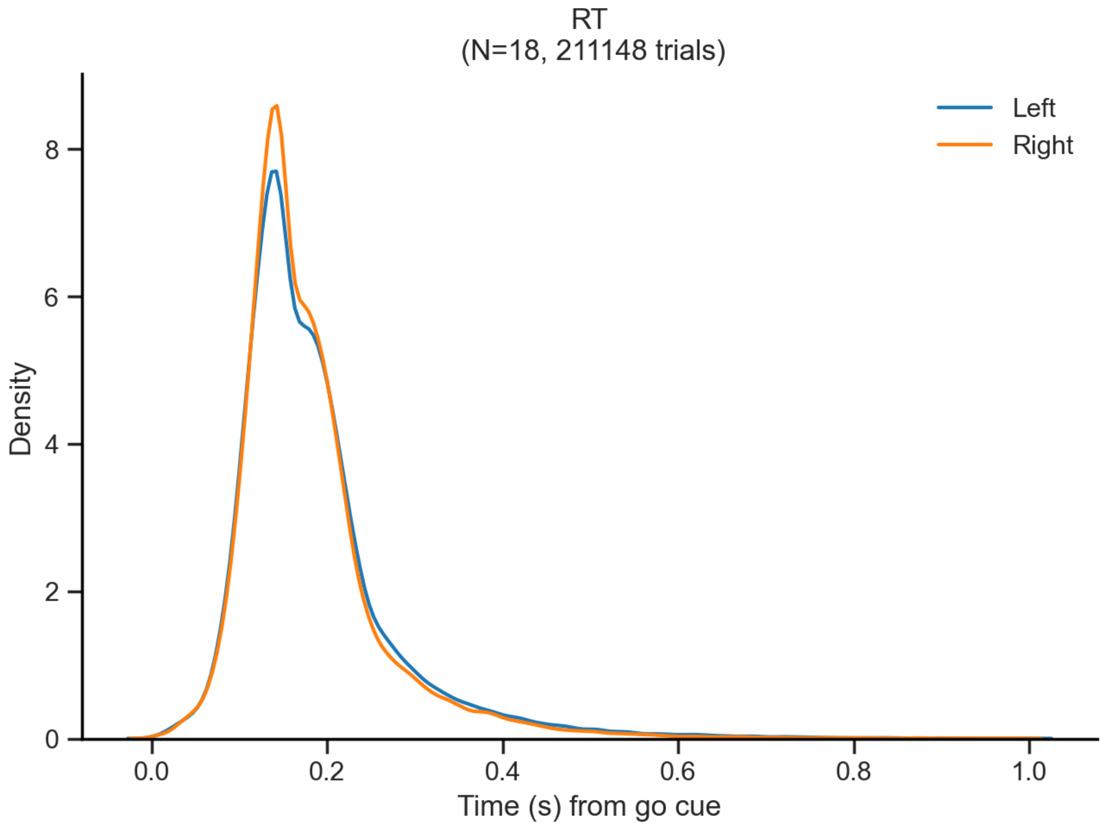
Repeat choice



Repeat trial Trials in which the stimulus (rewarded) side coincides with the previous animal's choice

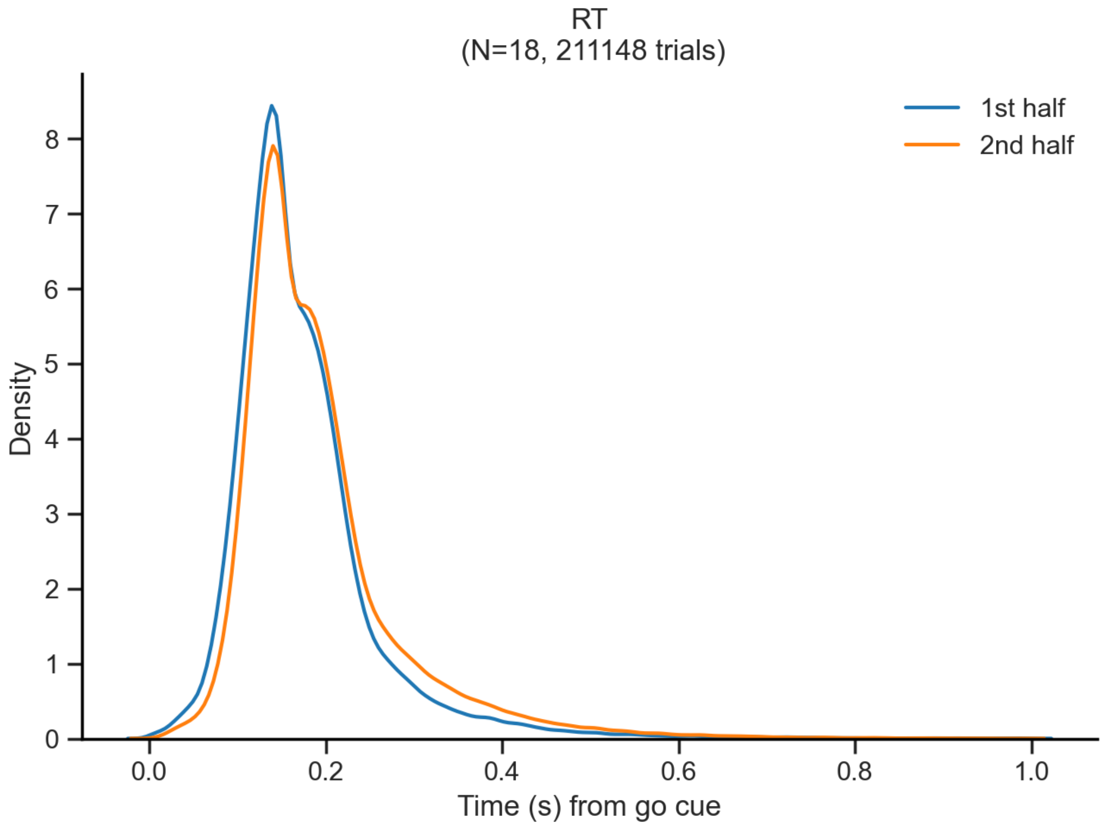

RT
(N=18, 211148 trials)
Alt.
Rep.
Density
Time (s) from go cue
0
1
2
3
4
5
6
7
8
0.0
0.2
0.4
0.6
0.8
1.0

Stimulus



This bimodality can not be explained by differences in behavioral variables such as outcome, choice, stimulus, repeat, or previous outcome.

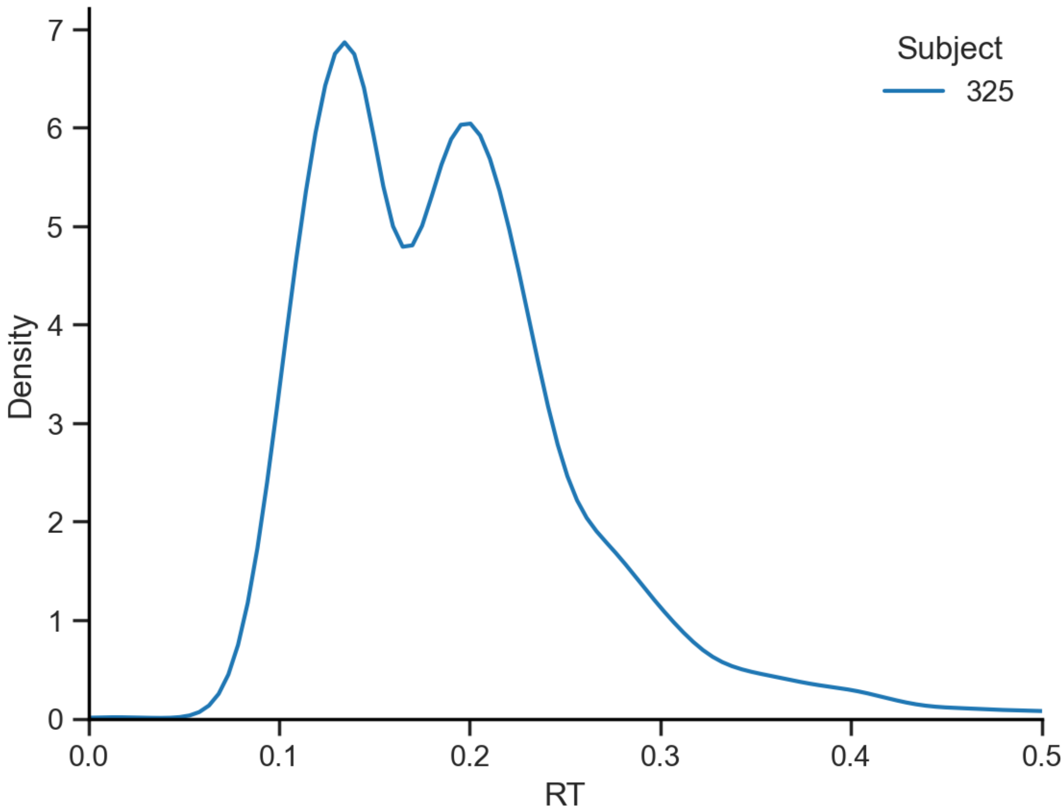
1st vs 2nd session half

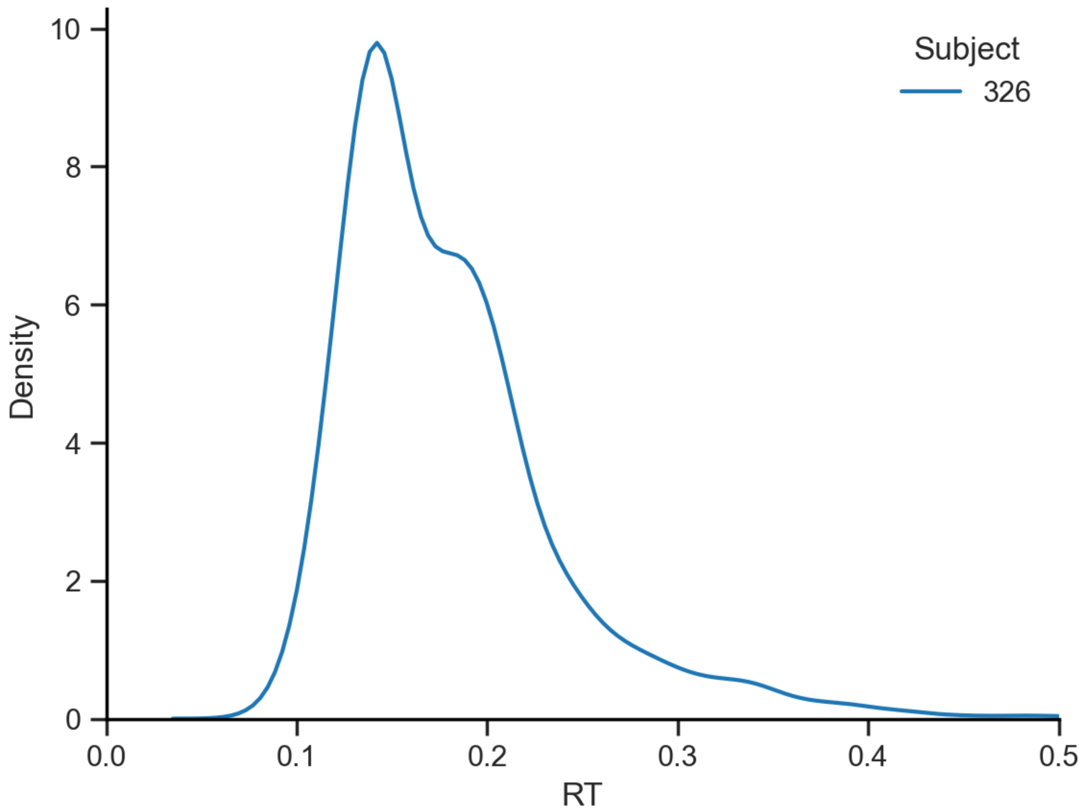


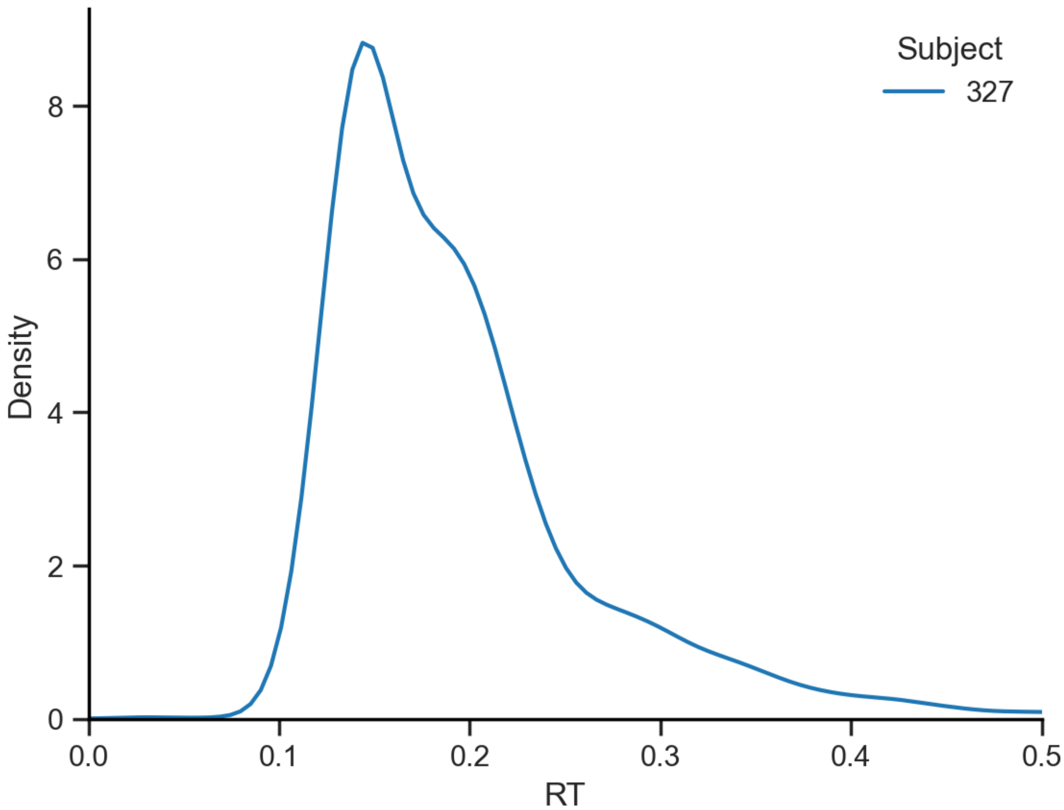
The bimodality is present in both halves of the session, suggesting that it is not driven by a change in the internal state of the animal. The RT distribution is similar in both halves of the session, with a slight shift towards slower RTs in the second half.

Another possibility is that the bimodality just results from the individual variability in the RTs of each subject. To examine this, we can plot the RT distribution for each subject separately. If the bimodality is driven by individual variability, we would expect to see a unimodal distribution in each subject.

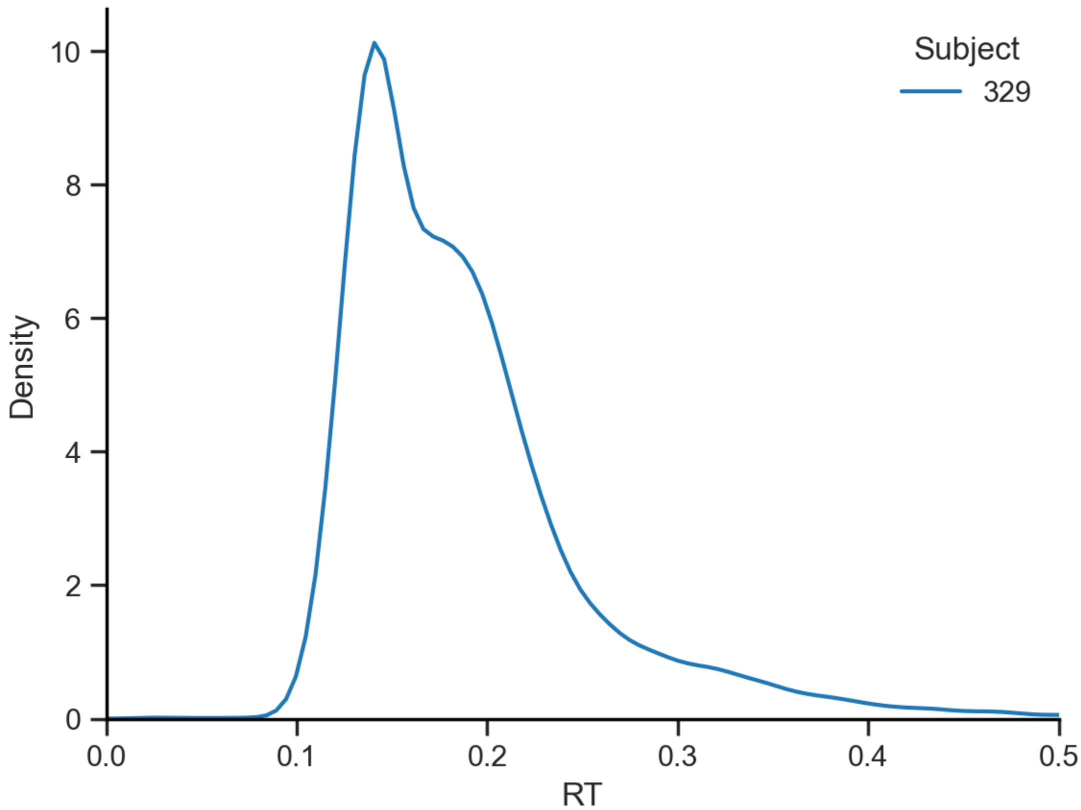
5.1.3 Check the bimodality is present in every subject

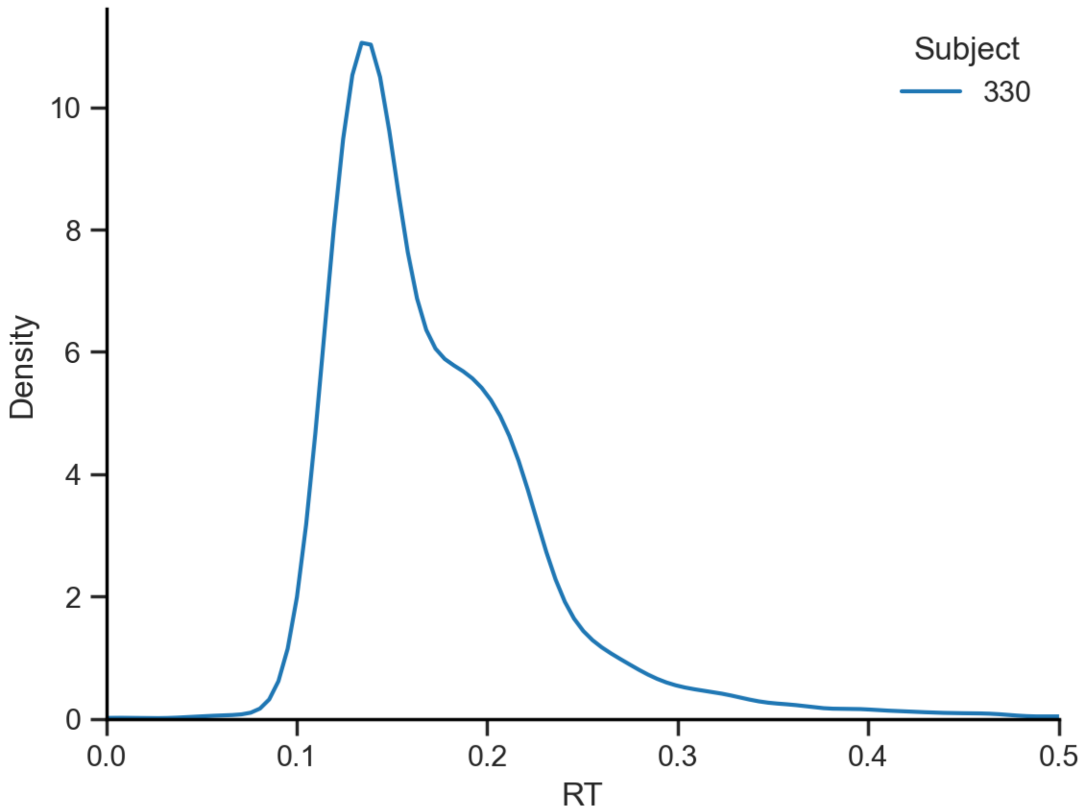


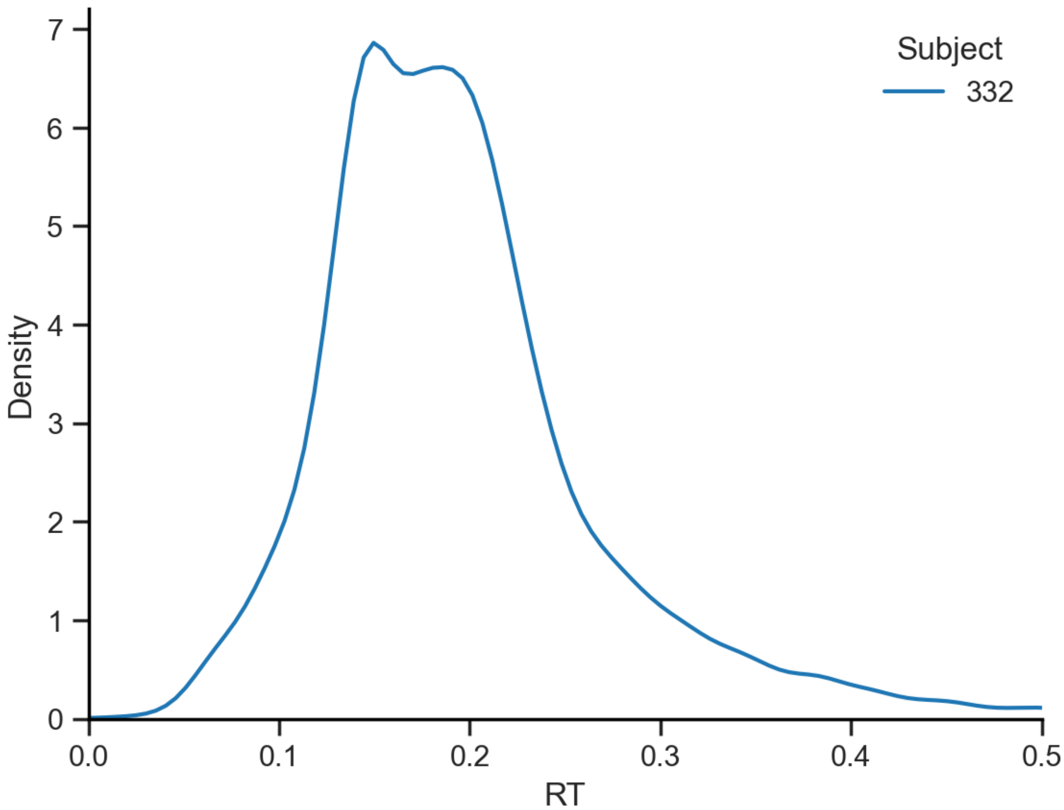

Subject
326
Density
RT
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5

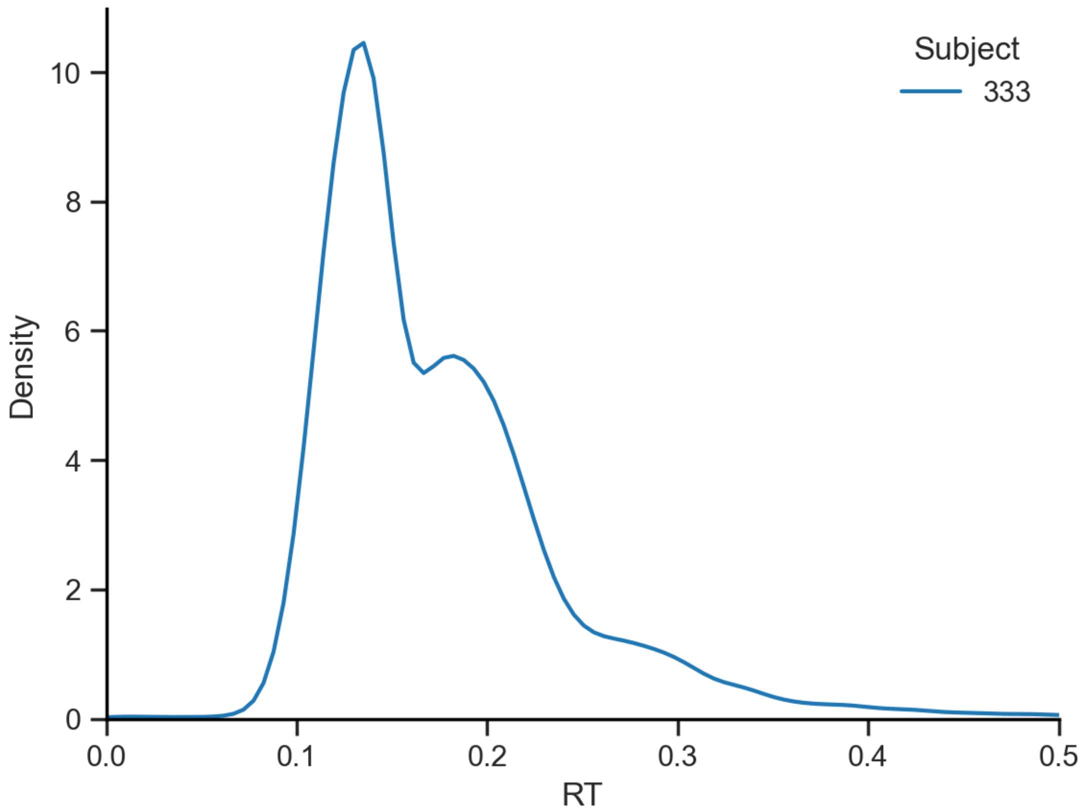


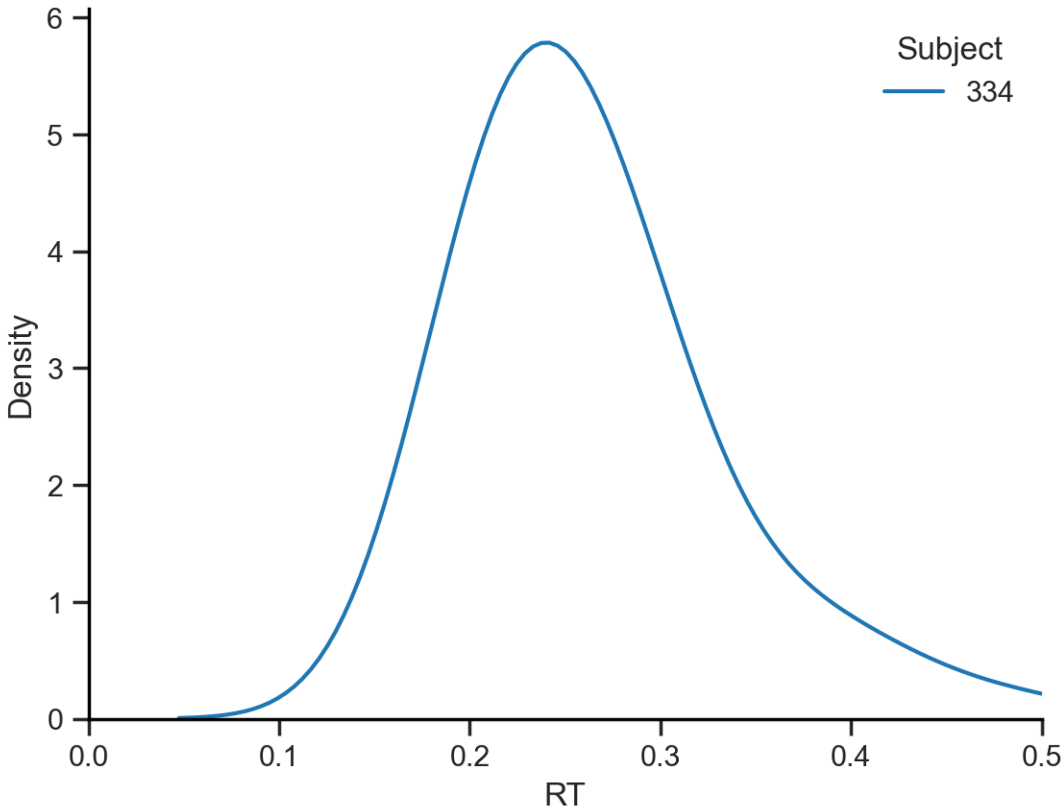
Subject
327
Density
RT
0
2
4
6
8
0.0
0.1
0.2
0.3
0.4
0.5

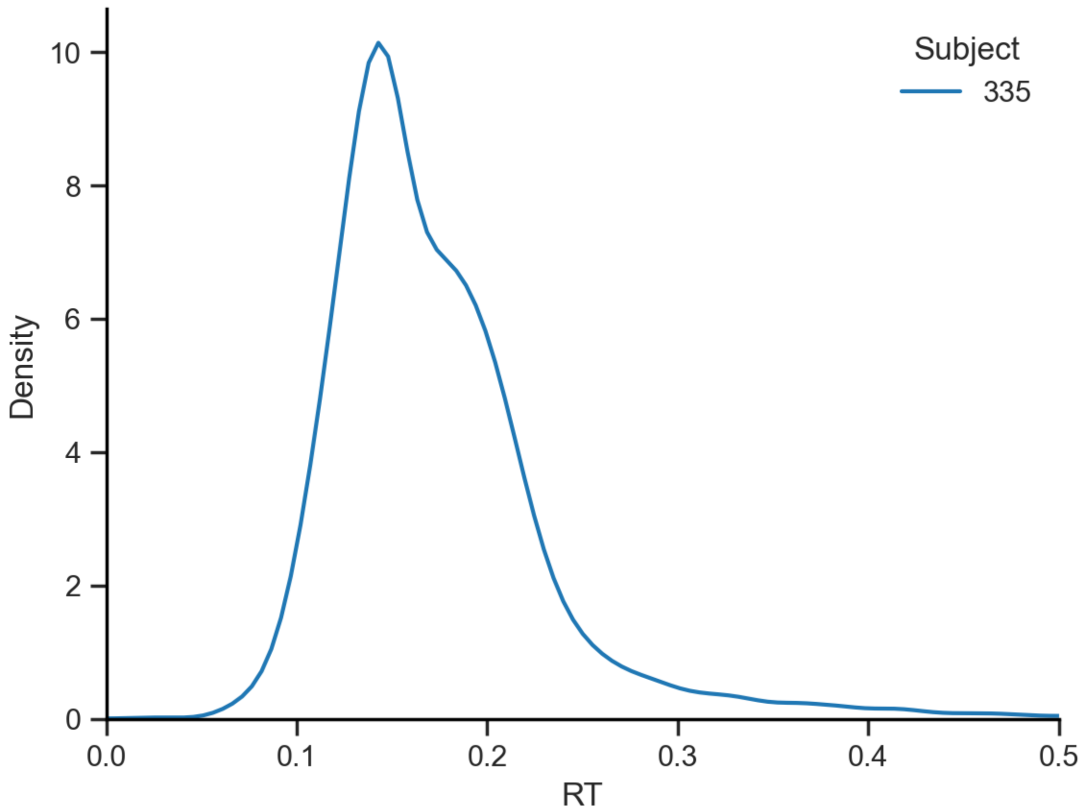

Subject
329
Density
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5
RT


Subject
330
Density
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5
RT

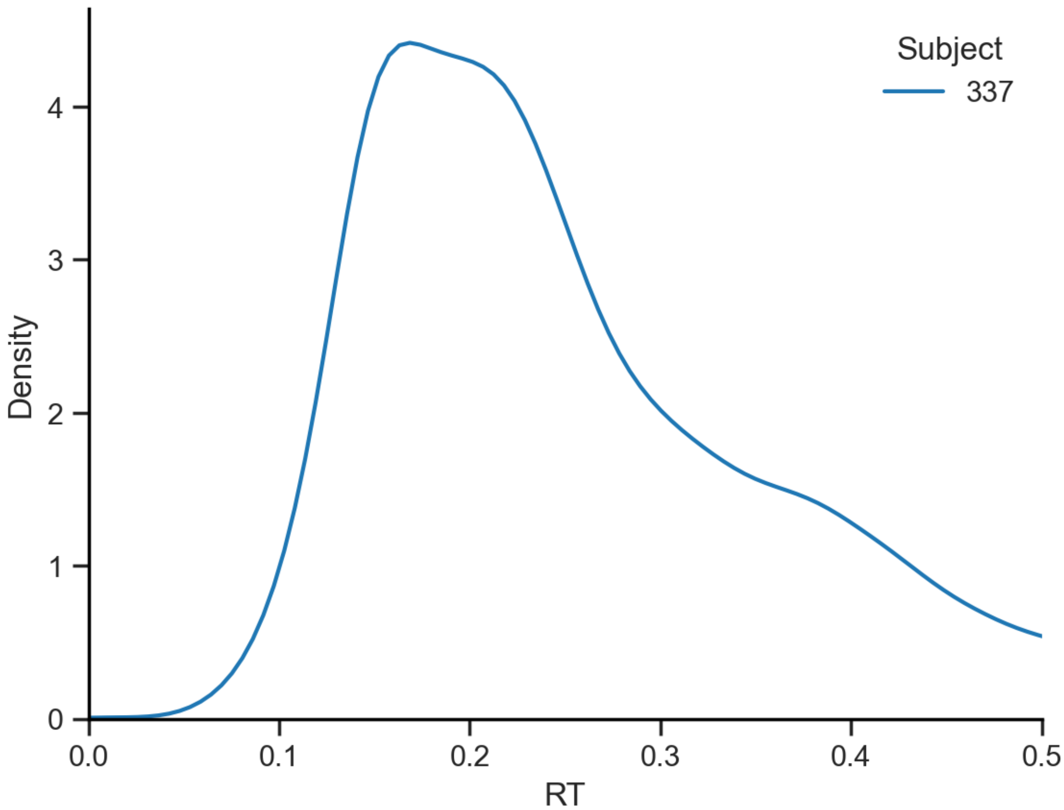

Subject
332
Density
RT
0
1
2
3
4
5
6
7
0.0
0.1
0.2
0.3
0.4
0.5

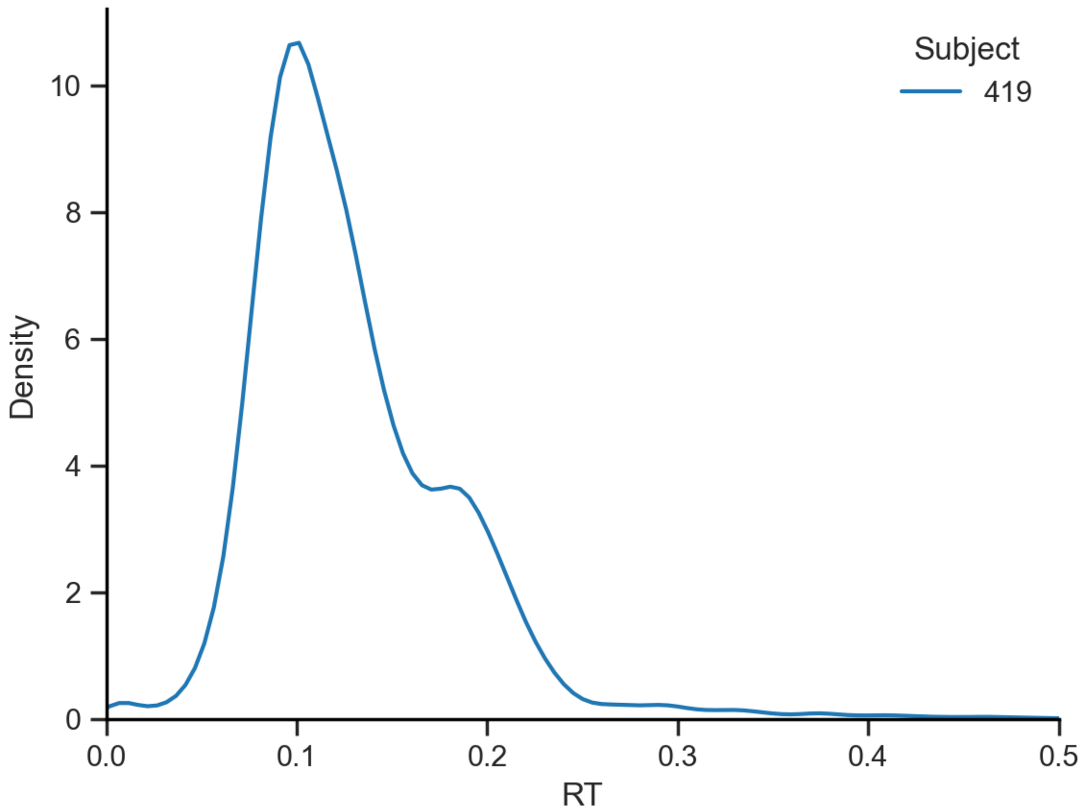

Subject
333
Density
RT
0
2
4
6
8
10
0.0
0.1
0.2
0.3
0.4
0.5

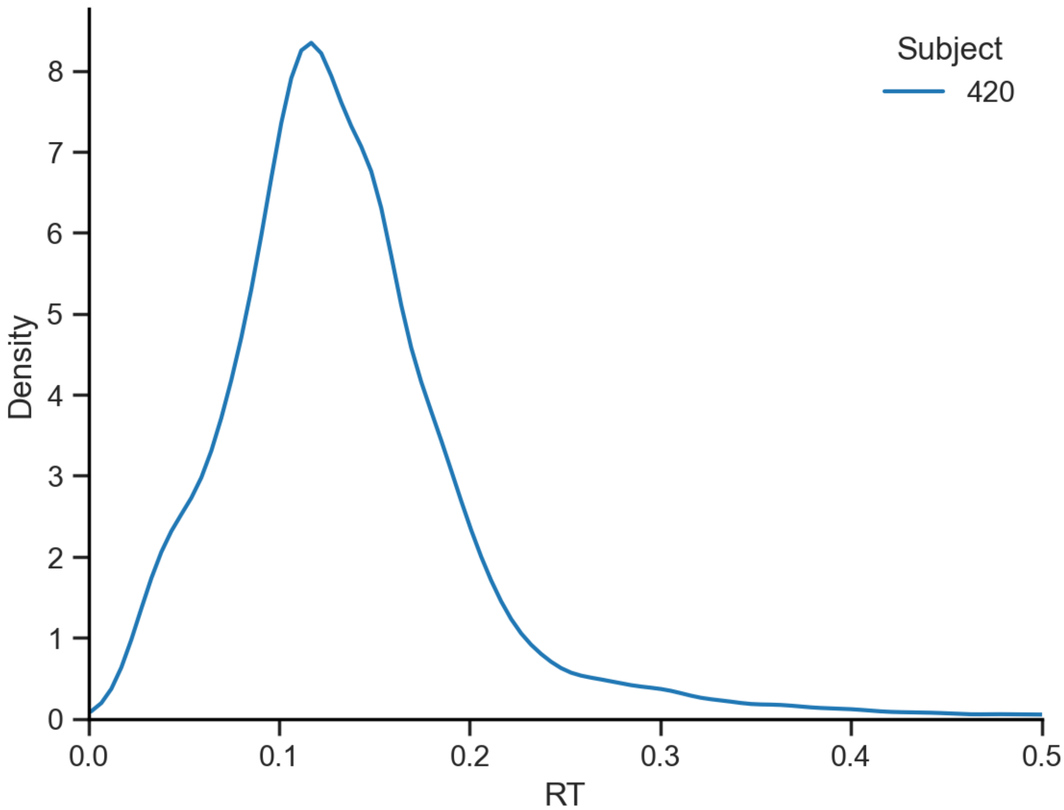

Subject
334
Density
RT
0
1
2
3
4
5
6
0.0
0.1
0.2
0.3
0.4
0.5



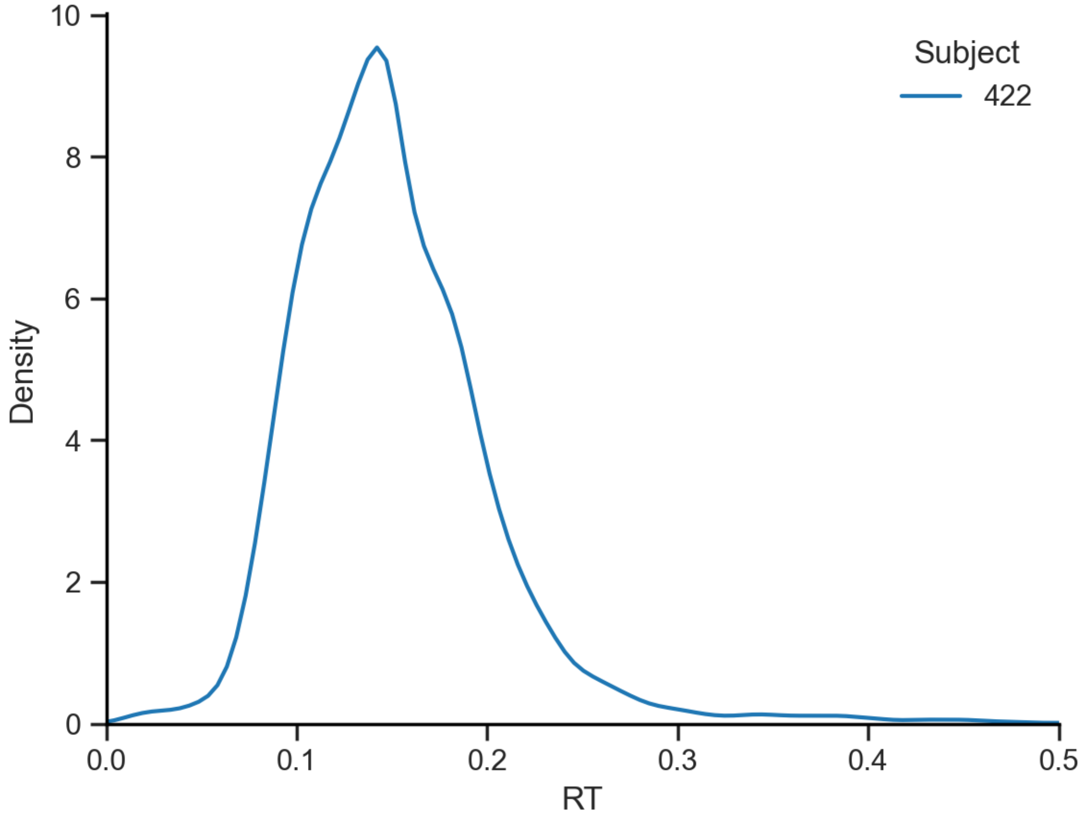
Subject
335
Density
RT
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5

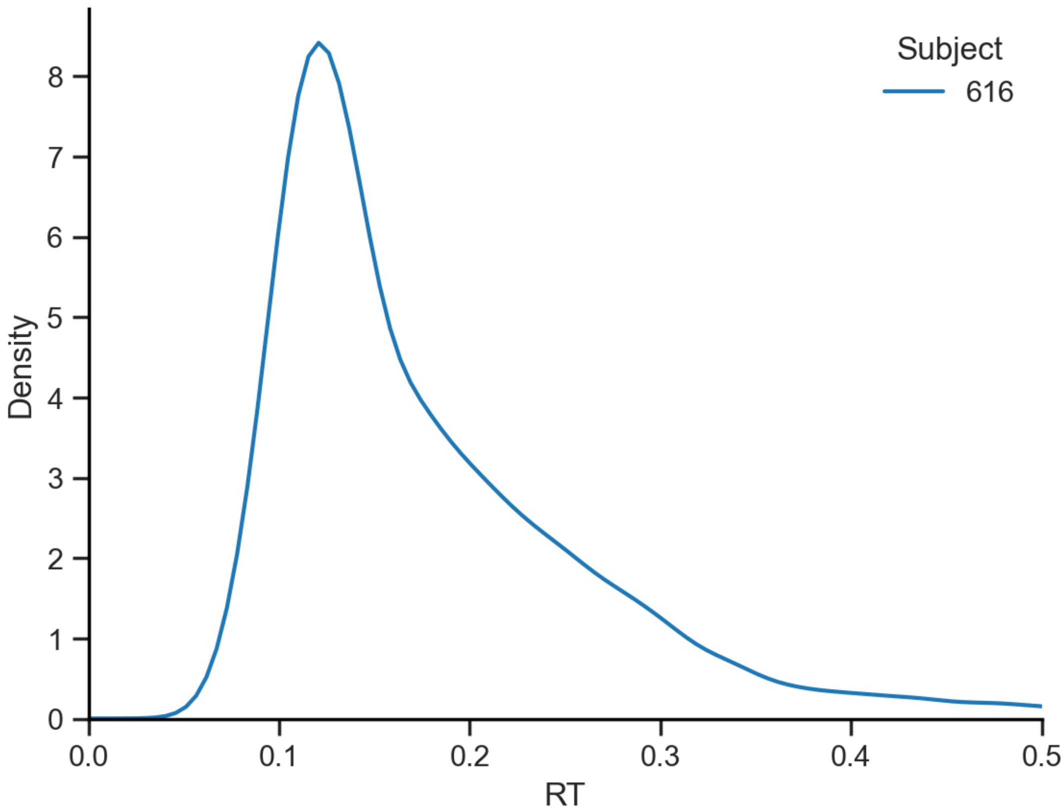

Subject
337
Density
4
3
2
1
0
0.0
0.1
0.2
0.3
0.4
0.5
RT


Subject
419
Density
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5
RT

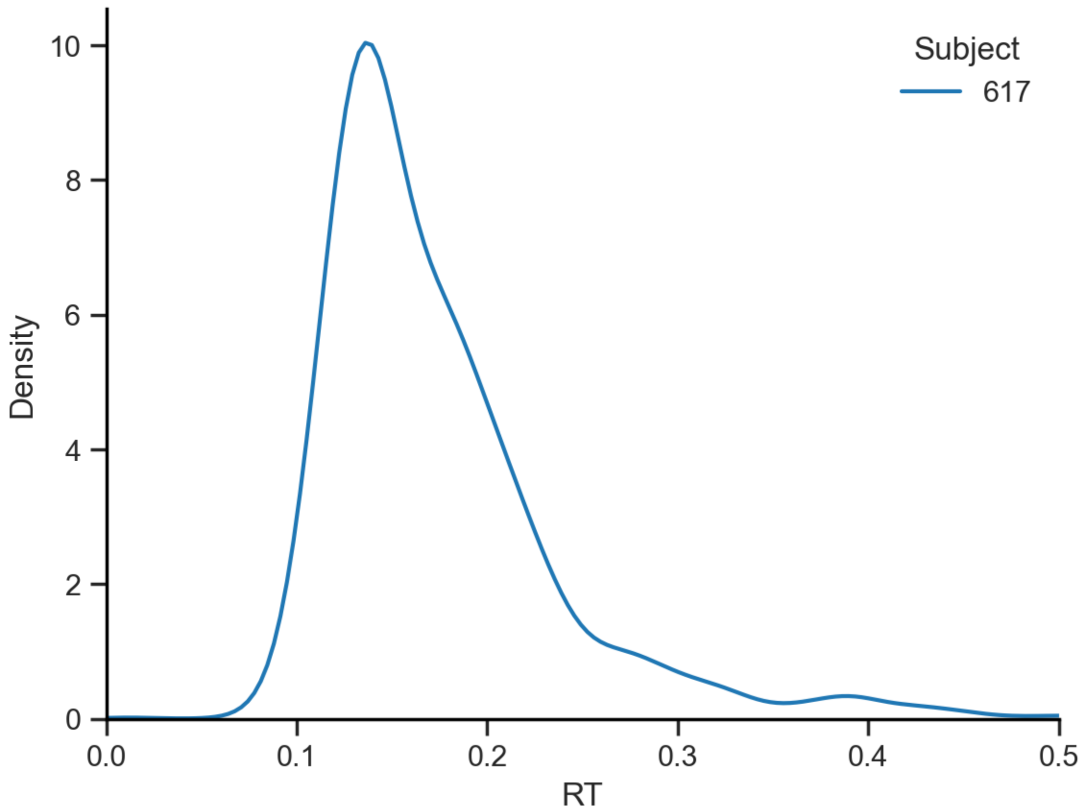


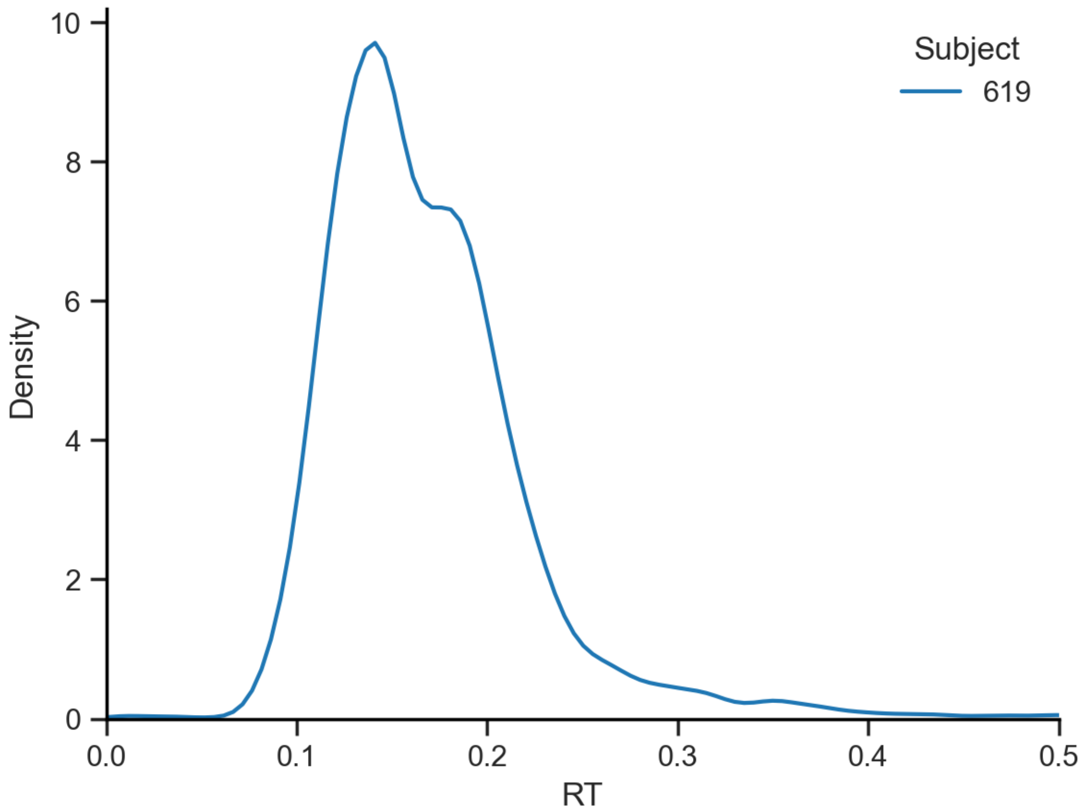
Subject
420
Density
RT
0
1
2
3
4
5
6
7
8
0.0
0.1
0.2
0.3
0.4
0.5


Subject
422
Density
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5
RT

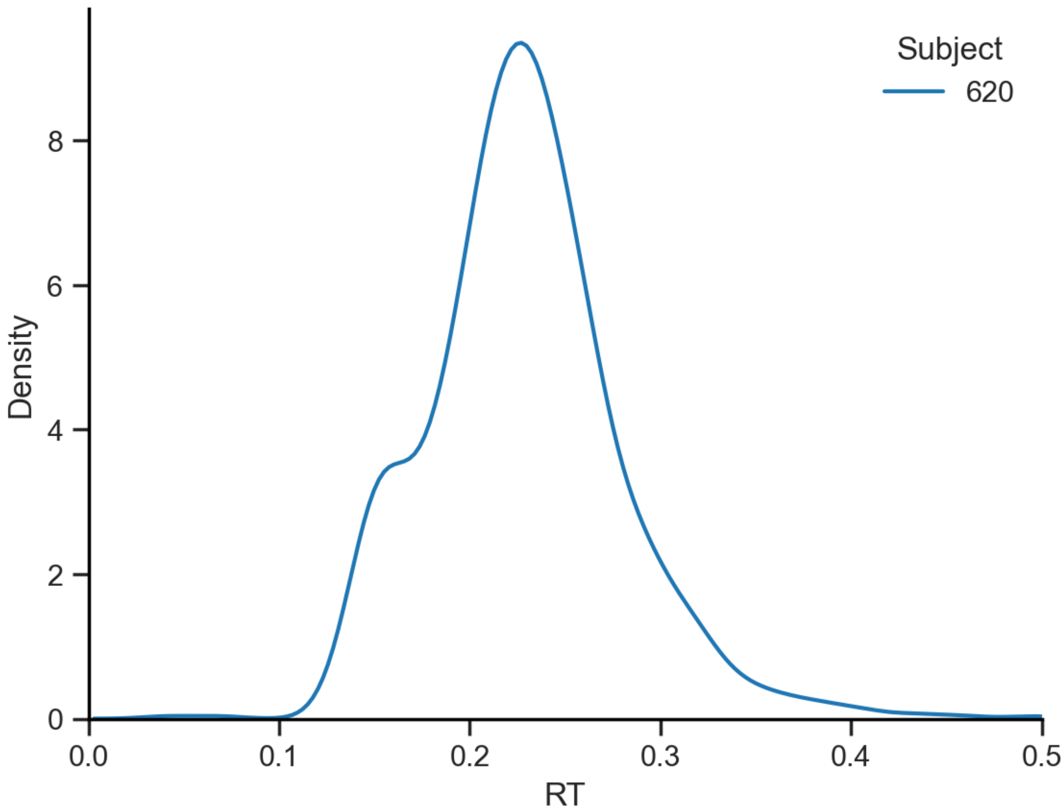


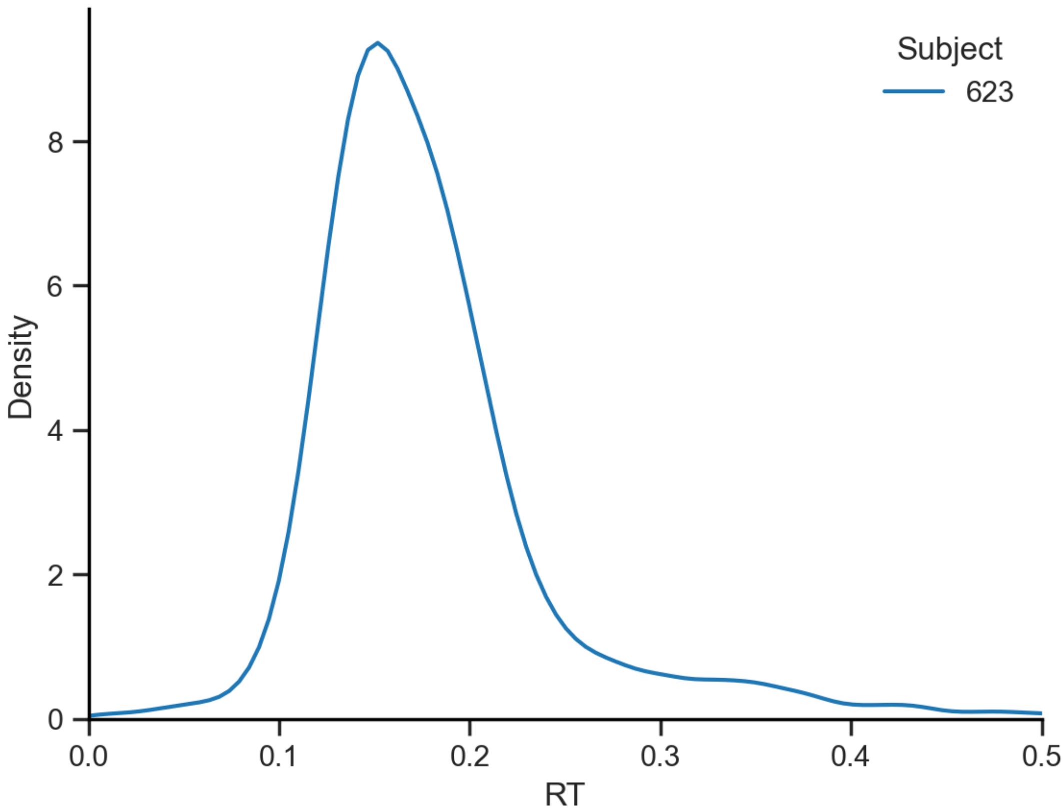
Subject
616
Density
RT
0
1
2
3
4
5
6
7
8
0.0
0.1
0.2
0.3
0.4
0.5


Subject
617
Density
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5
RT



Subject
619
Density
RT
10
8
6
4
2
0
0.0
0.1
0.2
0.3
0.4
0.5


Subject
620
Density
RT
0
2
4
6
8
0.0
0.1
0.2
0.3
0.4
0.5



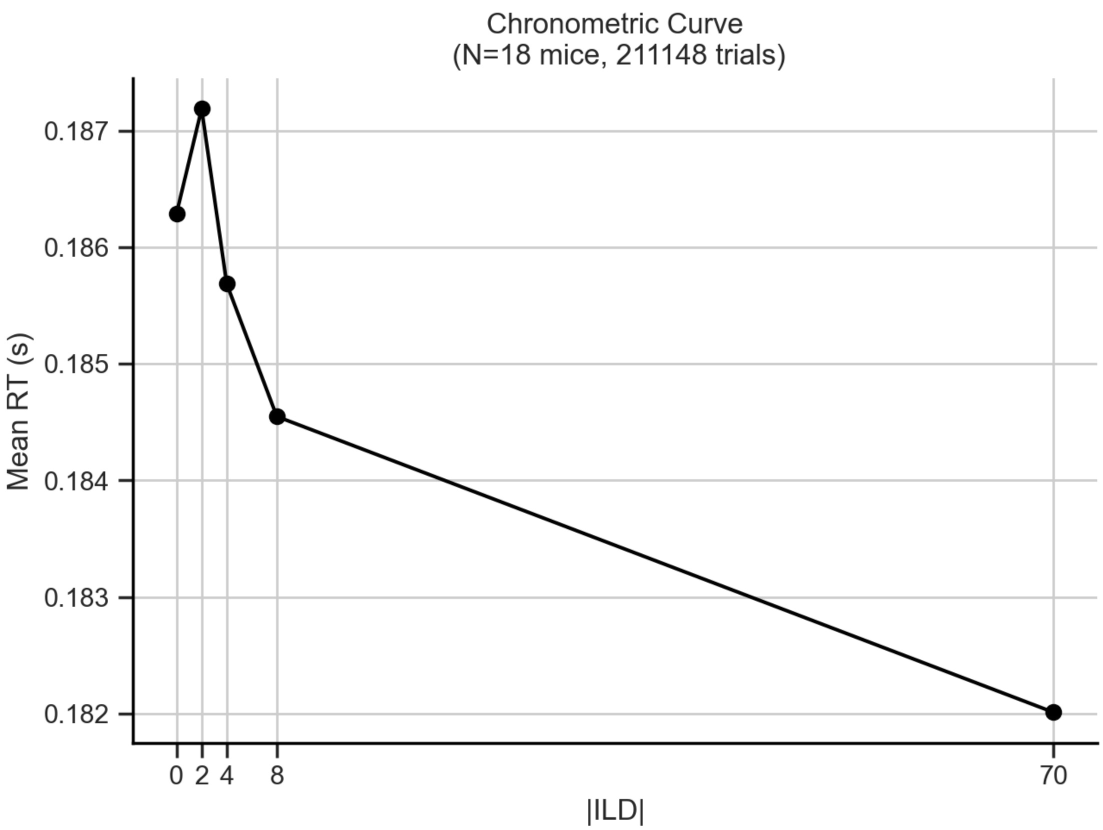
5.1.4 Hidden Markov Model (HMM)

Another possibility is that the bimodality reflects the a change in the internal state of the animal (e.g. motivation, reduced by satiation as time passes) that is not captured by the behavioral variables. To examine this, we can split the data of each session into first and second half and plot the RT distribution for each half. If the bimodality is driven by a change in the internal state of the animal, we would expect to see a difference in the RT distribution between the first and second half of the session.

5.2 Chronometric Curves

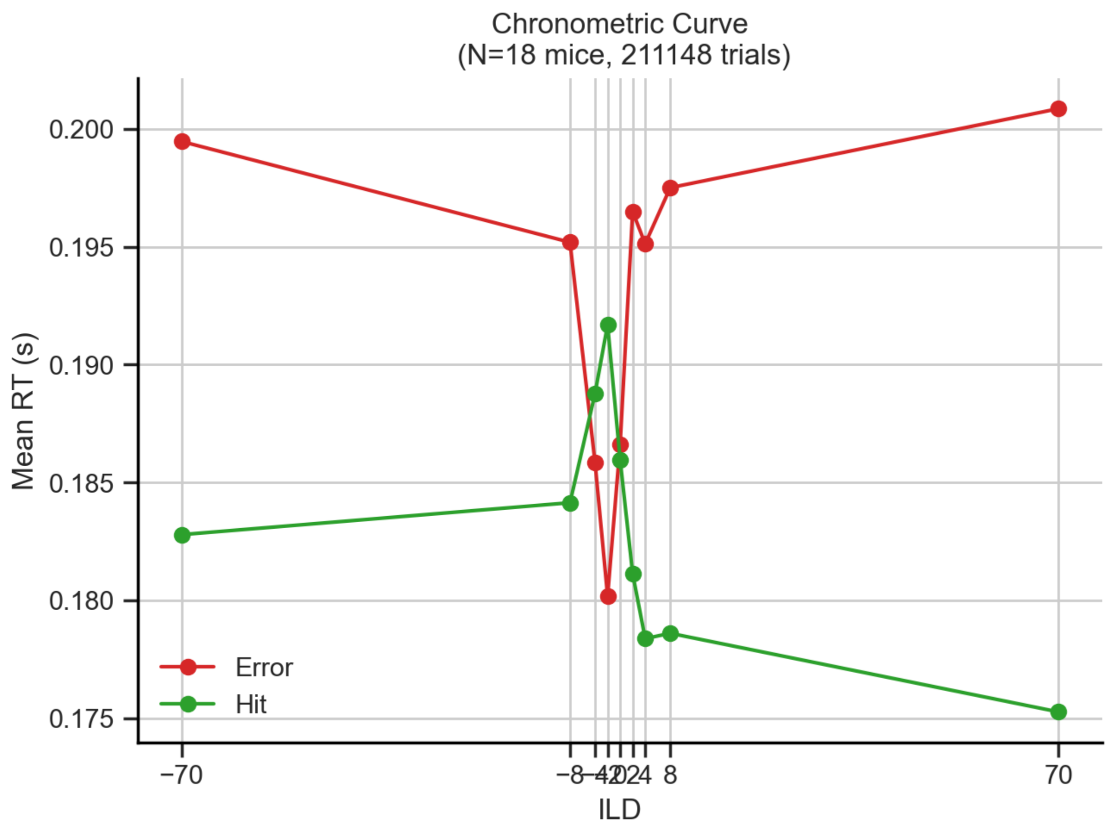
5.2.1 All trials

```
   absILD        RT
0     0.0  0.186289
1     2.0  0.187186
2     4.0  0.185685
3     8.0  0.184547
4    70.0  0.182010
```



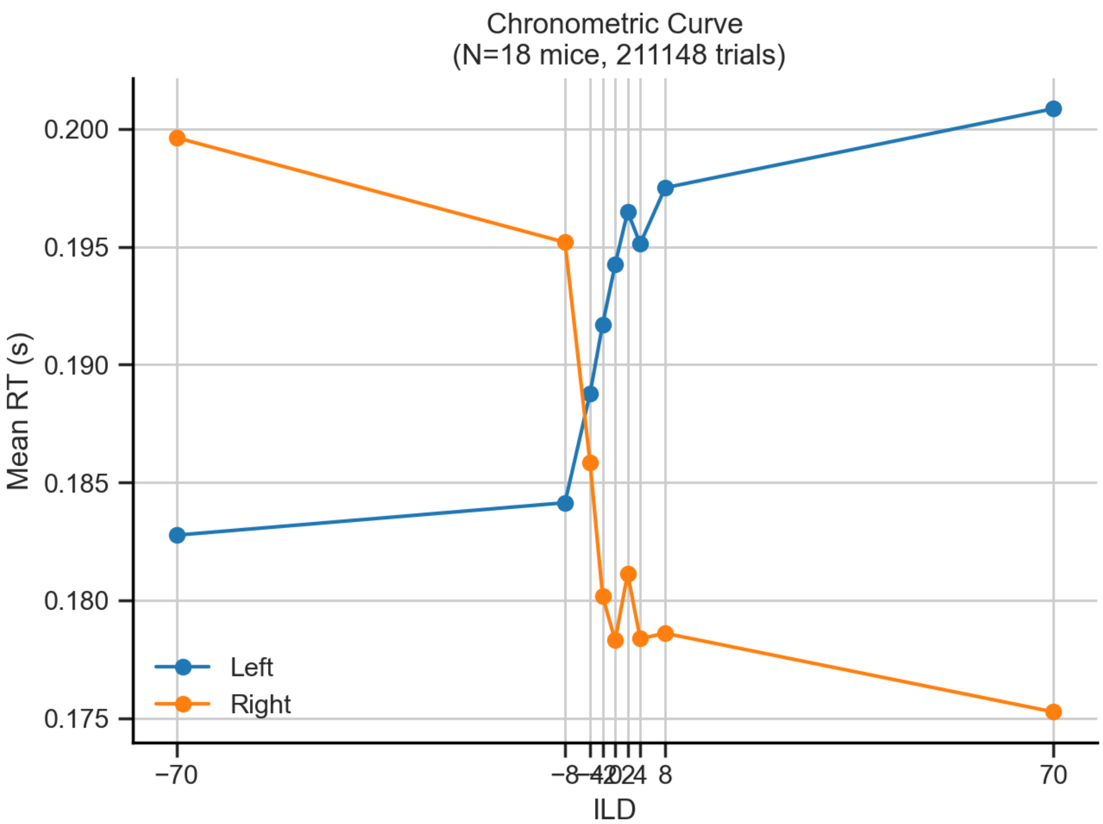
5.2.2 Splits

Outcome

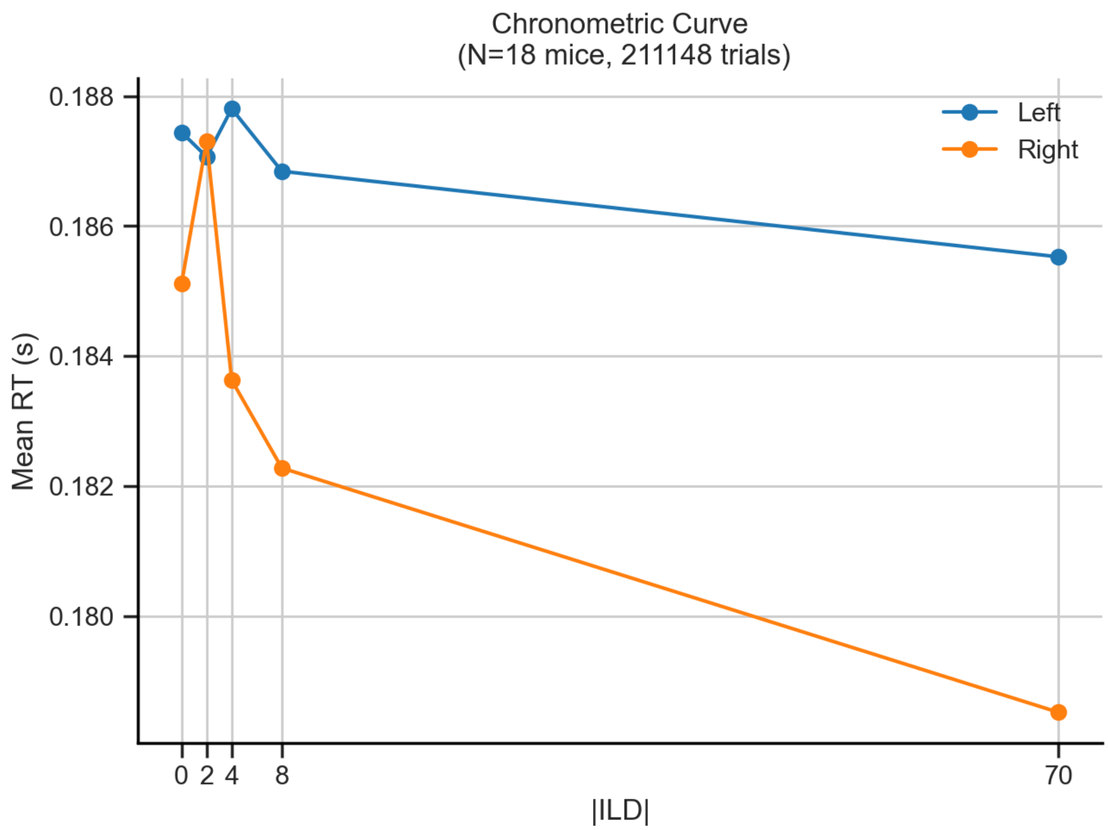


Chronometric Curve
(N=18 mice, 211148 trials)
Mean RT (s)
0.200
0.195
0.190
0.185
0.180
0.175
Error
Hit
−70
−8
−4
−2
0
2
4
8
70
ILD

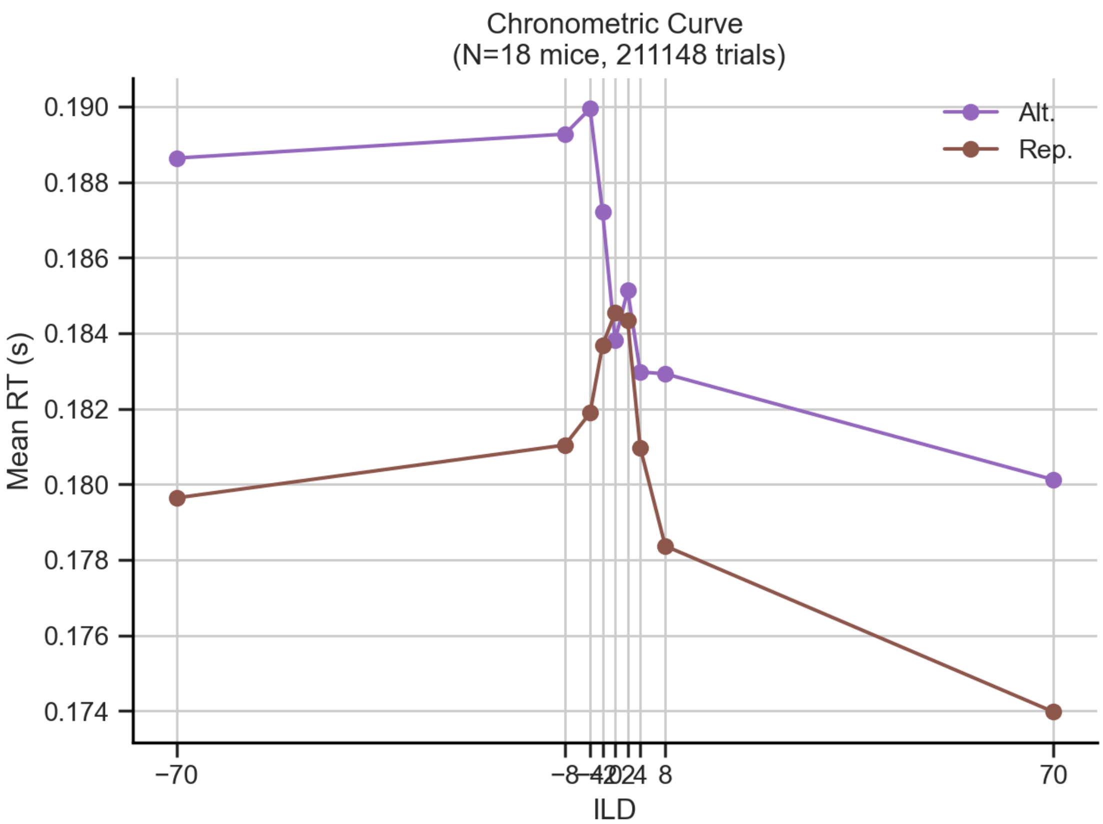
Choice


Chronometric Curve
(N=18 mice, 211148 trials)
Mean RT (s)
0.200
0.195
0.190
0.185
0.180
0.175
Left
Right
−70
−8
−4
−2
0
2
4
8
70
ILD

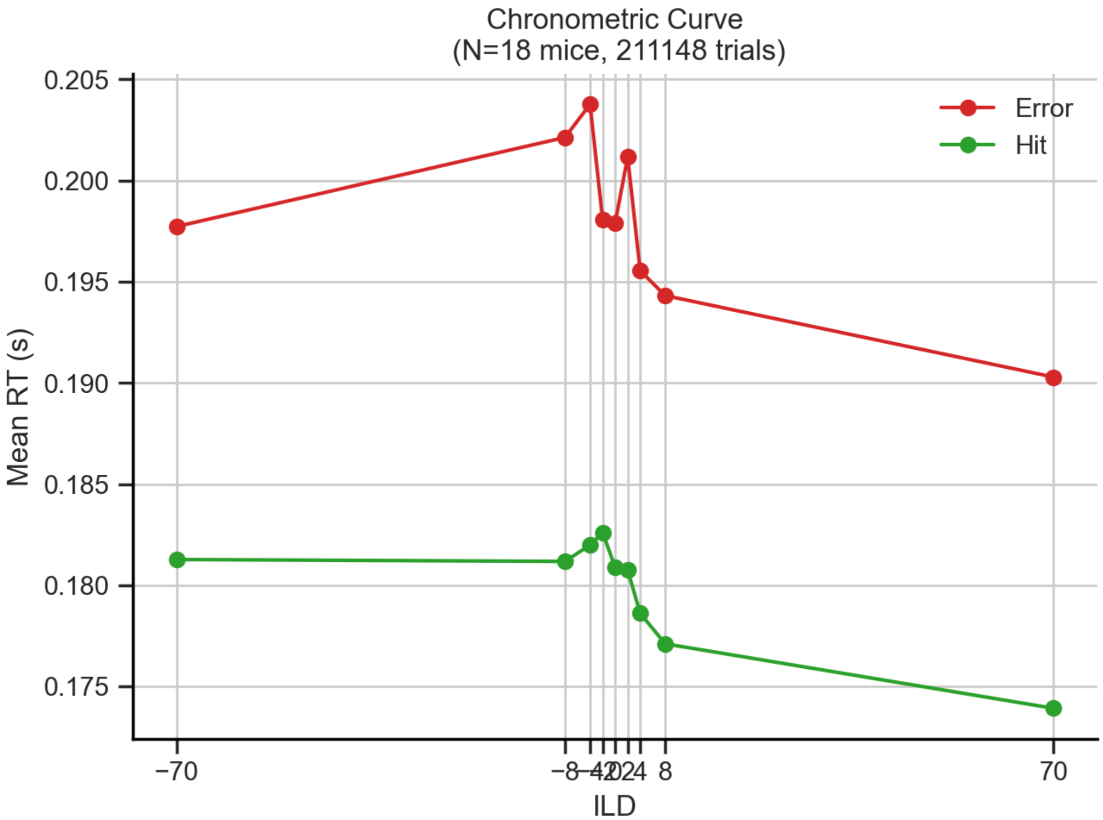
Stimulus


Chronometric Curve
(N=18 mice, 211148 trials)
Mean RT (s)
0.188
0.186
0.184
0.182
0.180
|ILD|
0
2
4
8
70
Left
Right

Repeat



Previous outcome



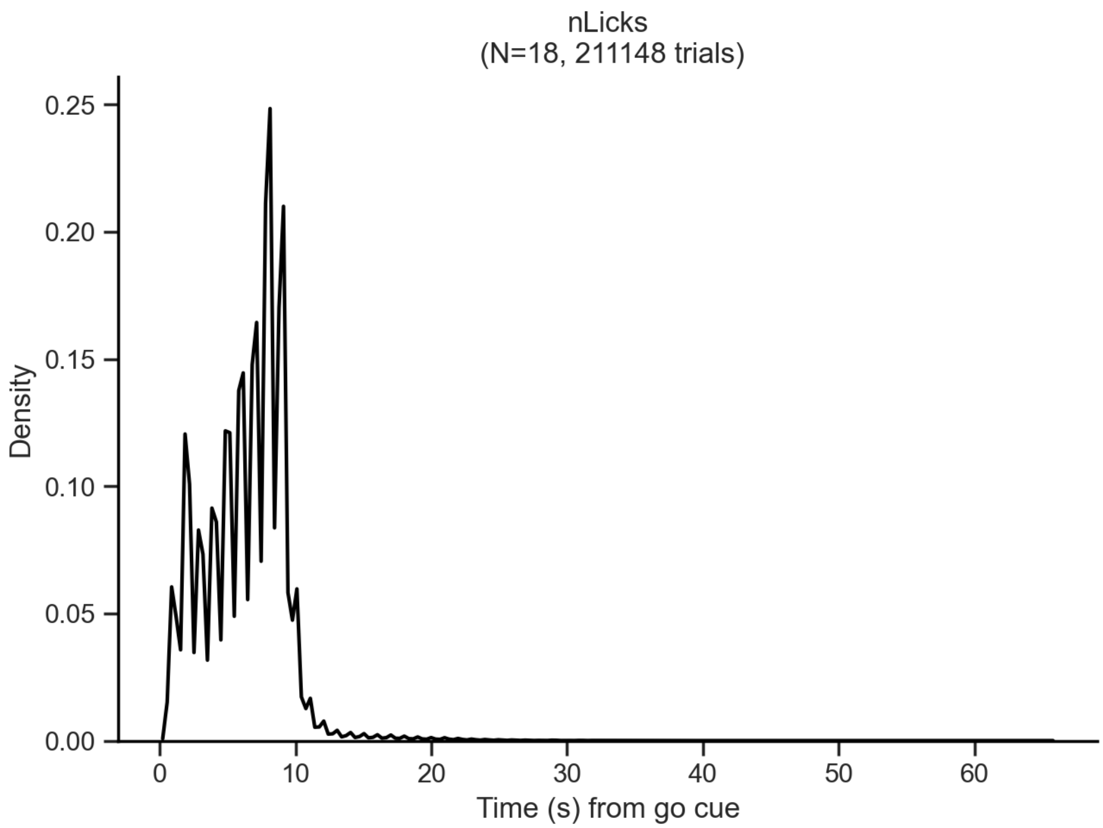
6 Lick Rate

As the number of licks depends on the time the lickport was available (motor in), which ultimately depends on the outcome (correct vs error), a time window to count the licks needed to be established. As the time to collect water in correct (rewarded) trials was always 1 s, and the timeout period following error trials varied from 2-5 s, the chosen time window to count the number of licks was 1 s. This is very convenient for the lick rate, as it is defined as

$$\text{Lick rate} = \frac{N_{\text{licks}}}{\text{time (s)}}$$

so that N licks = Lick Rate

6.1 All trials

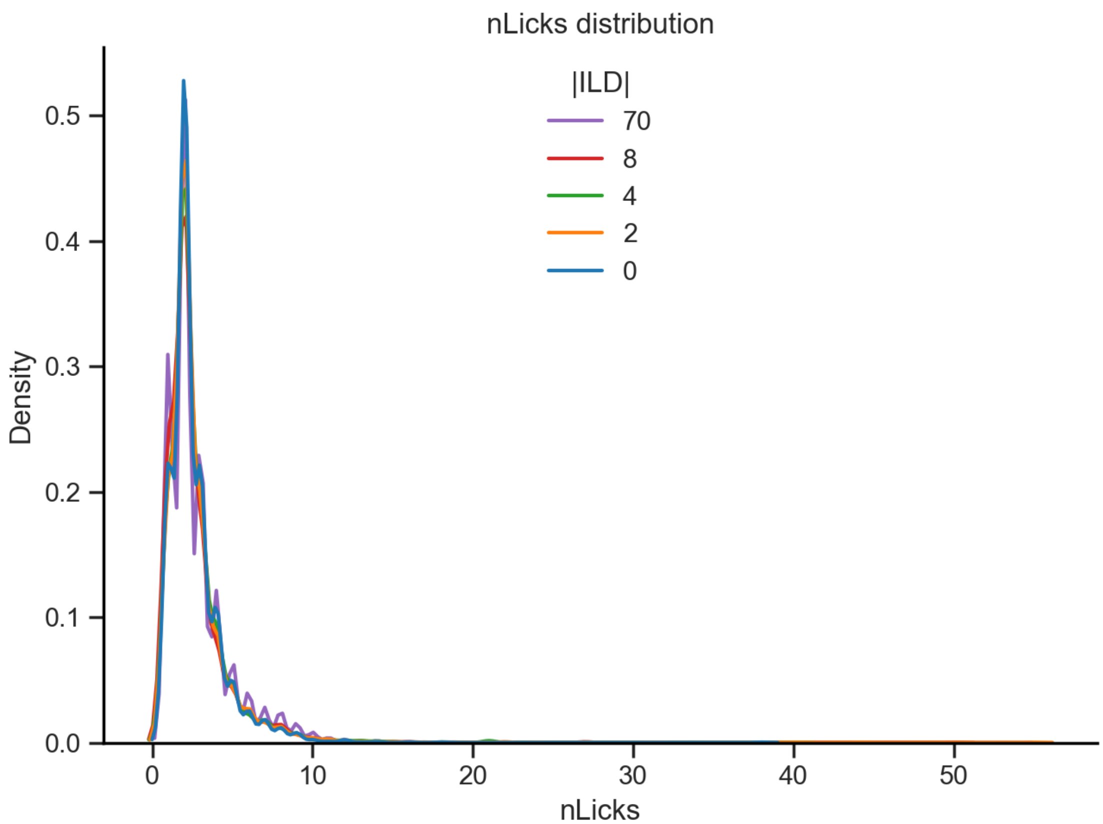


We can observe a first peak at 0 licks (error trials), and an sawtooth increasing second peak with max at 8 licks (correct trials). Let's split the data into several variables (like with RT) to observe how they modulate the lick rate.

6.2 Splits

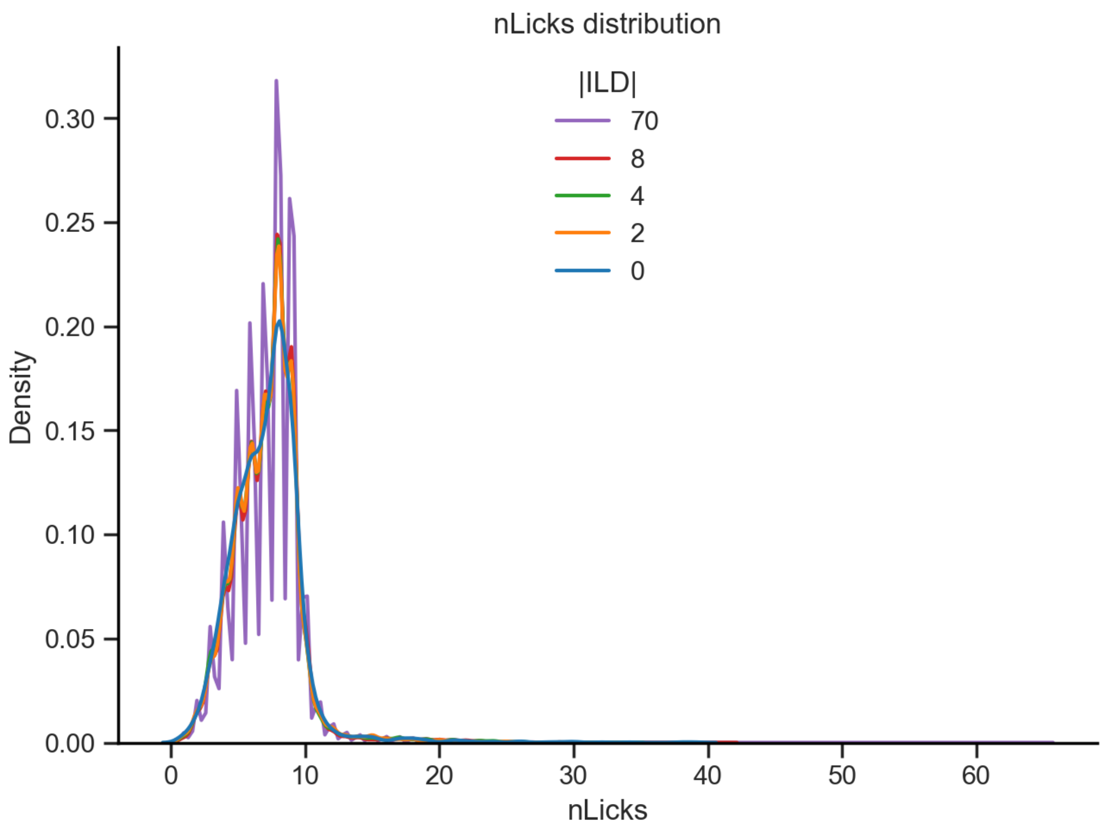
6.2.1 ILD

```
ILD 70: peak nLicks = 2.065 s
ILD 8: peak nLicks = 2.098 s
ILD 4: peak nLicks = 2.092 s
ILD 2: peak nLicks = 2.116 s
ILD 0: peak nLicks = 1.950 s
Mean nLicks = 2.064 s
```



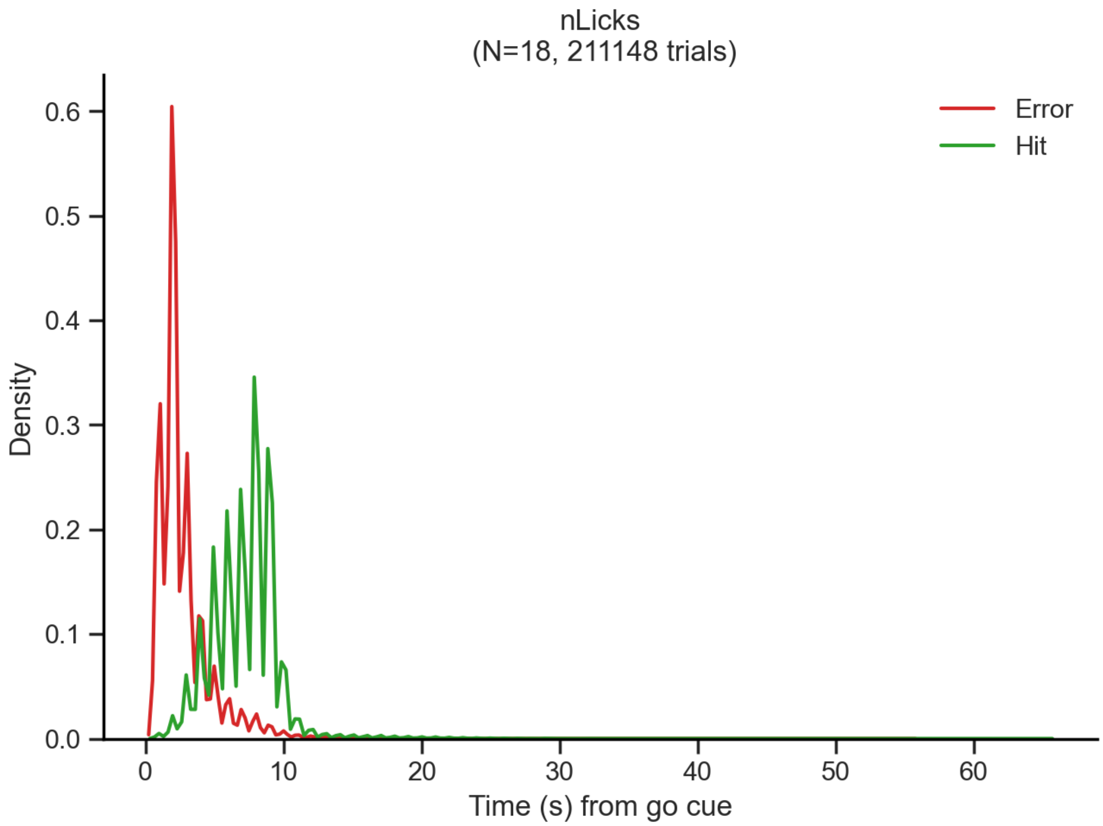
As there was no water (reward) coming out of the spouts in error trials, it makes sense that the distribution of lick rate peaks at 0. It is a needed control, but from now on the analysis will focus on **correct trials only**

```
ILD 70: peak nLicks = 7.861 s
ILD 8: peak nLicks = 7.919 s
ILD 4: peak nLicks = 7.982 s
ILD 2: peak nLicks = 8.053 s
ILD 0: peak nLicks = 8.094 s
Mean nLicks = 7.982 s
```

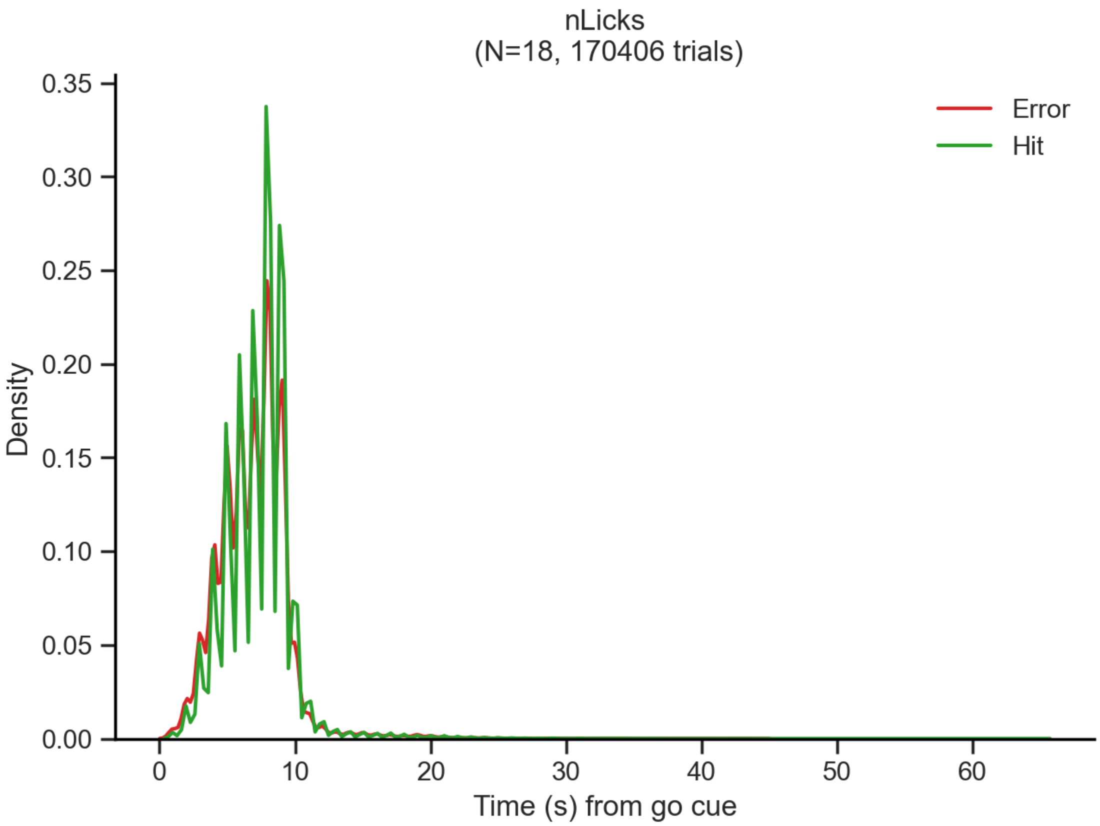


There is a modulation in the lick rate by ILD in correct trials. The higher the ILD the higher the lick rate.

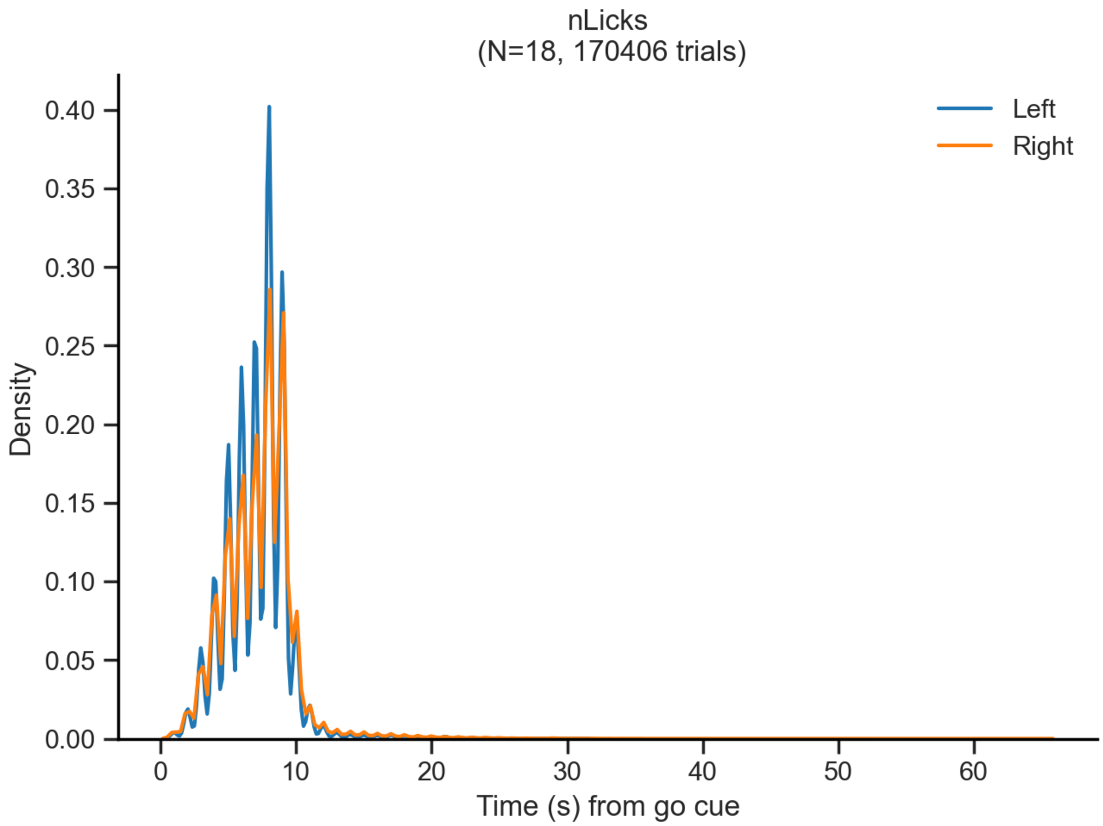
6.2.2 Outcome


nLicks
(N=18, 211148 trials)
Error
Hit
Density
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
10
20
30
40
50
60
Time (s) from go cue

6.2.3 Previous outcome

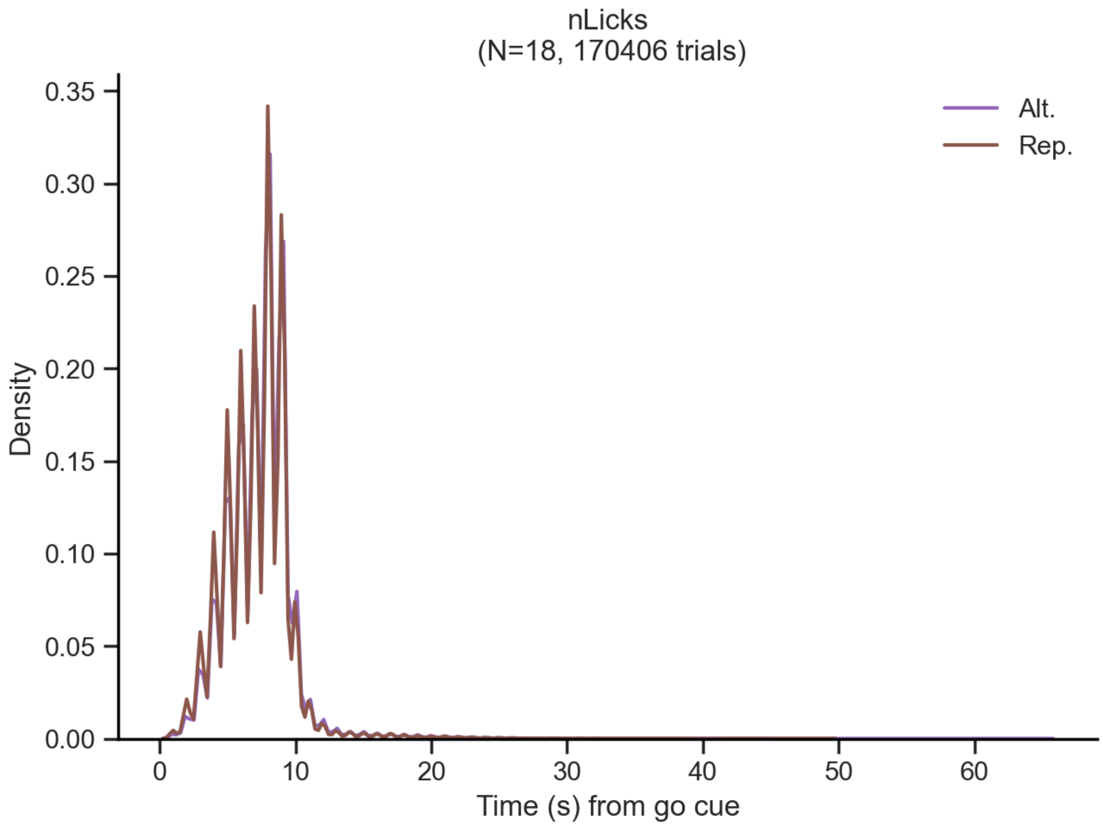


6.2.4 Choice

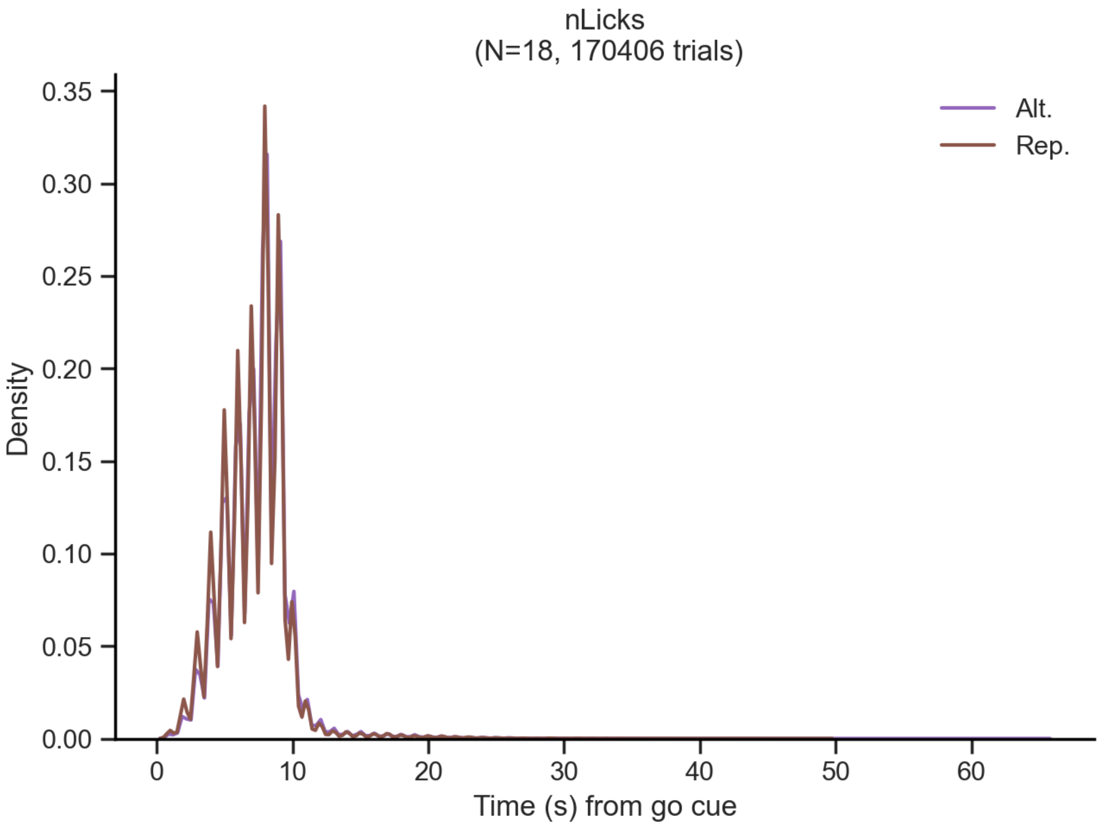


nLicks
(N=18, 170406 trials)
Left
Right
Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
40
50
60
Time (s) from go cue

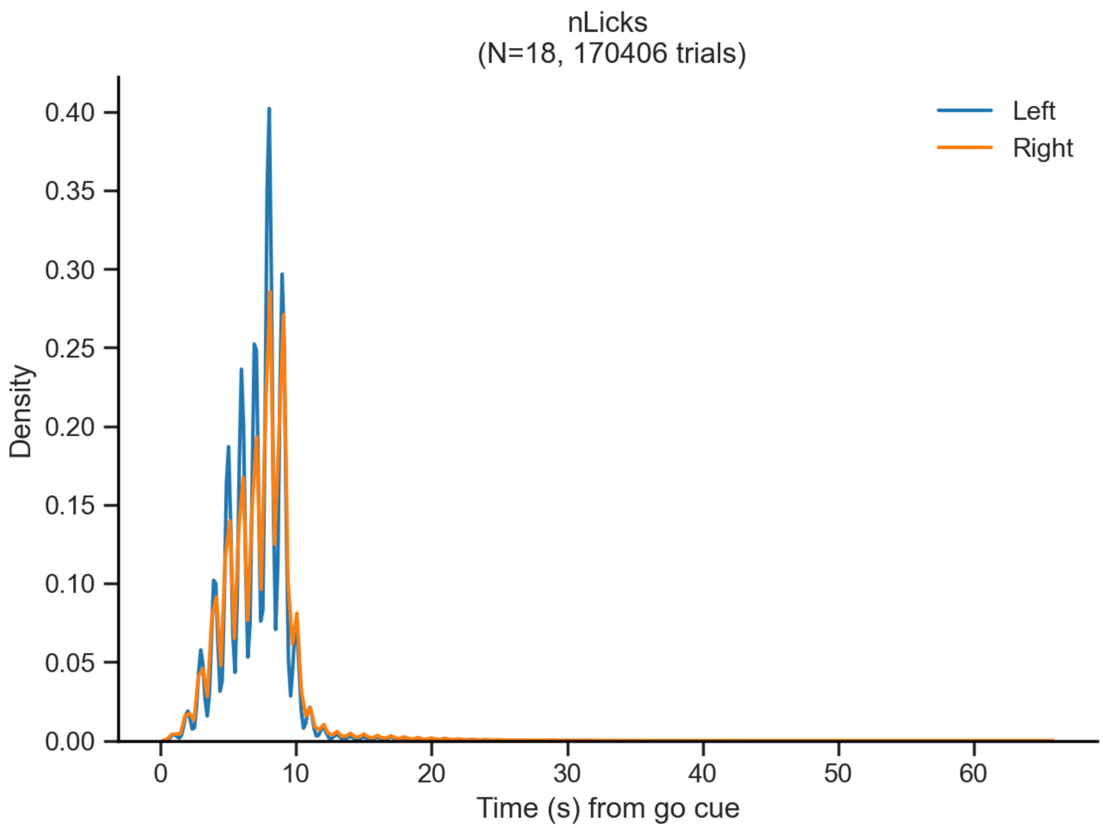
6.2.5 Repeat choice



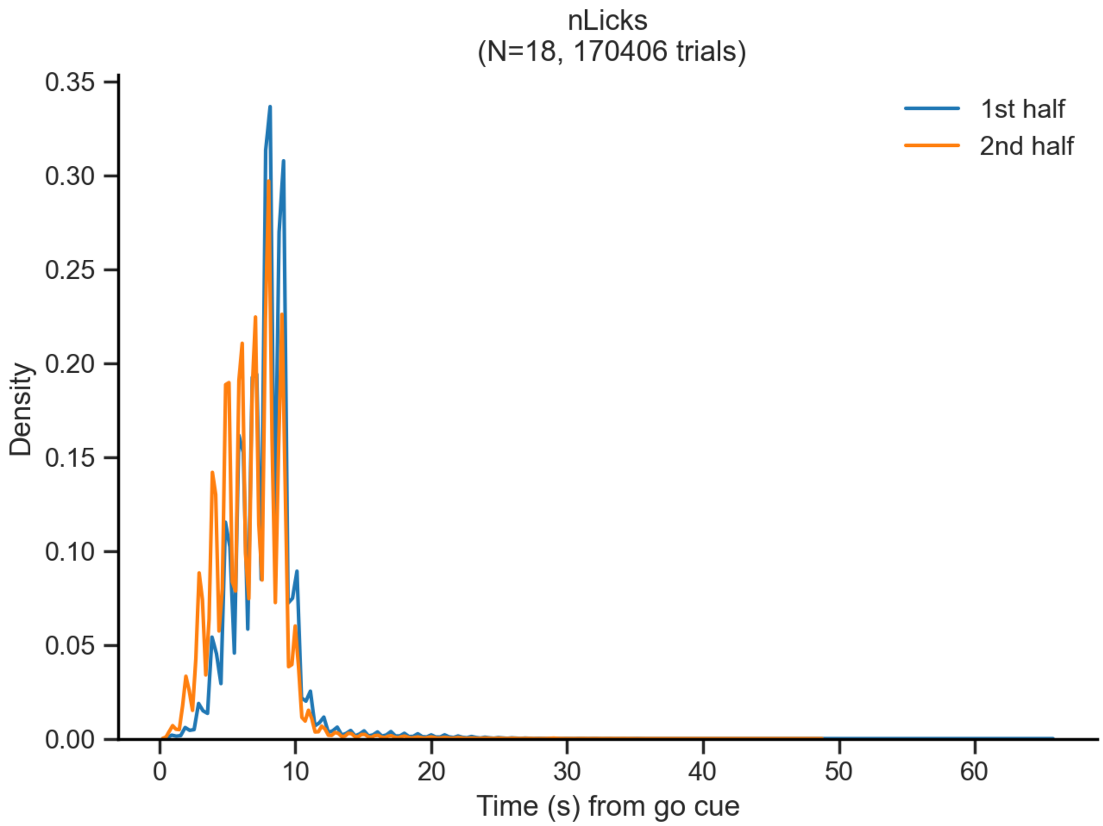
6.2.6 Repeat trial



6.2.7 Stimulus



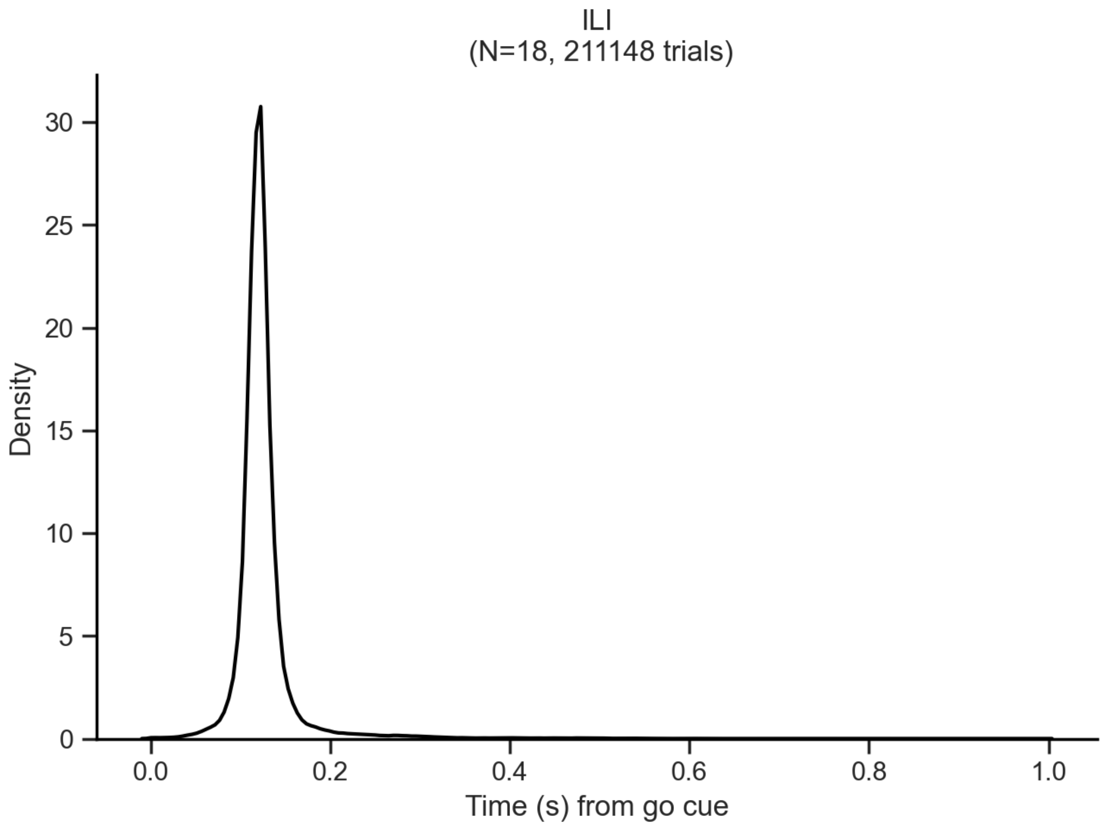
6.2.8 1st vs 2nd session half



7 Inter Lick Interval (ILI)

$$\text{Lick rate (Hz)} \approx \frac{1}{\text{Mean ILI (s)}}$$

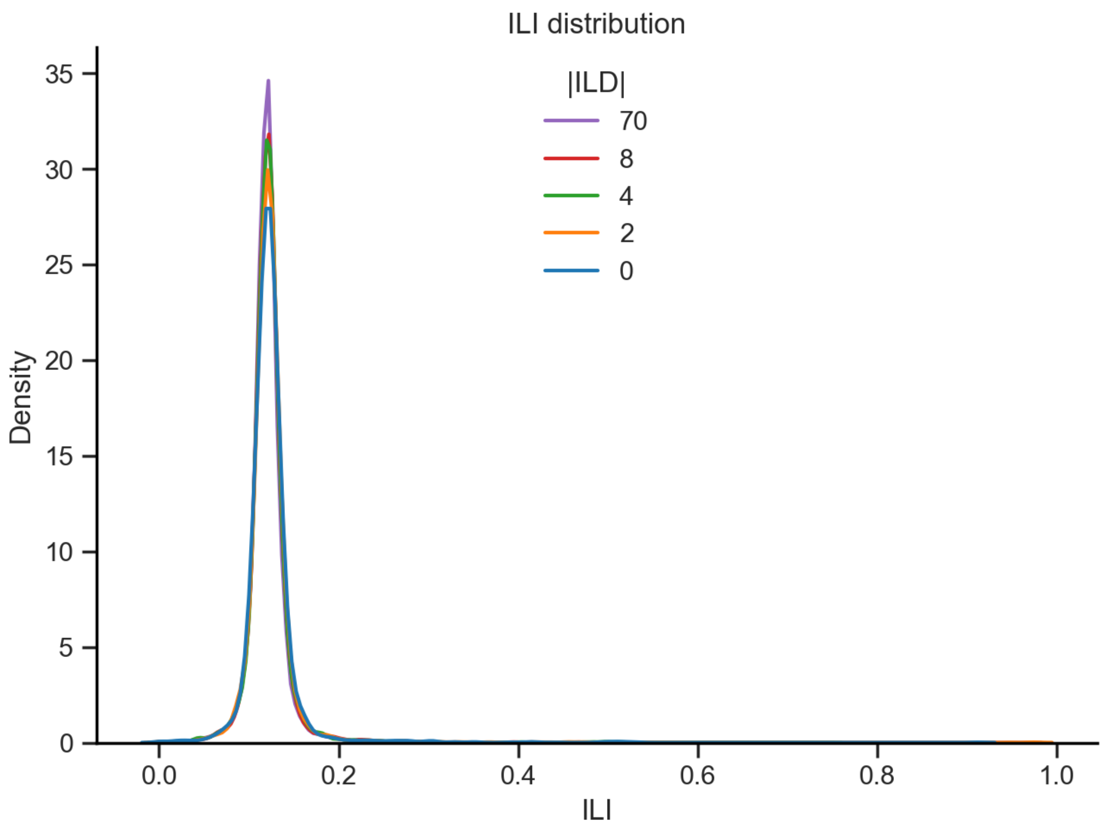
7.0.1 All trials



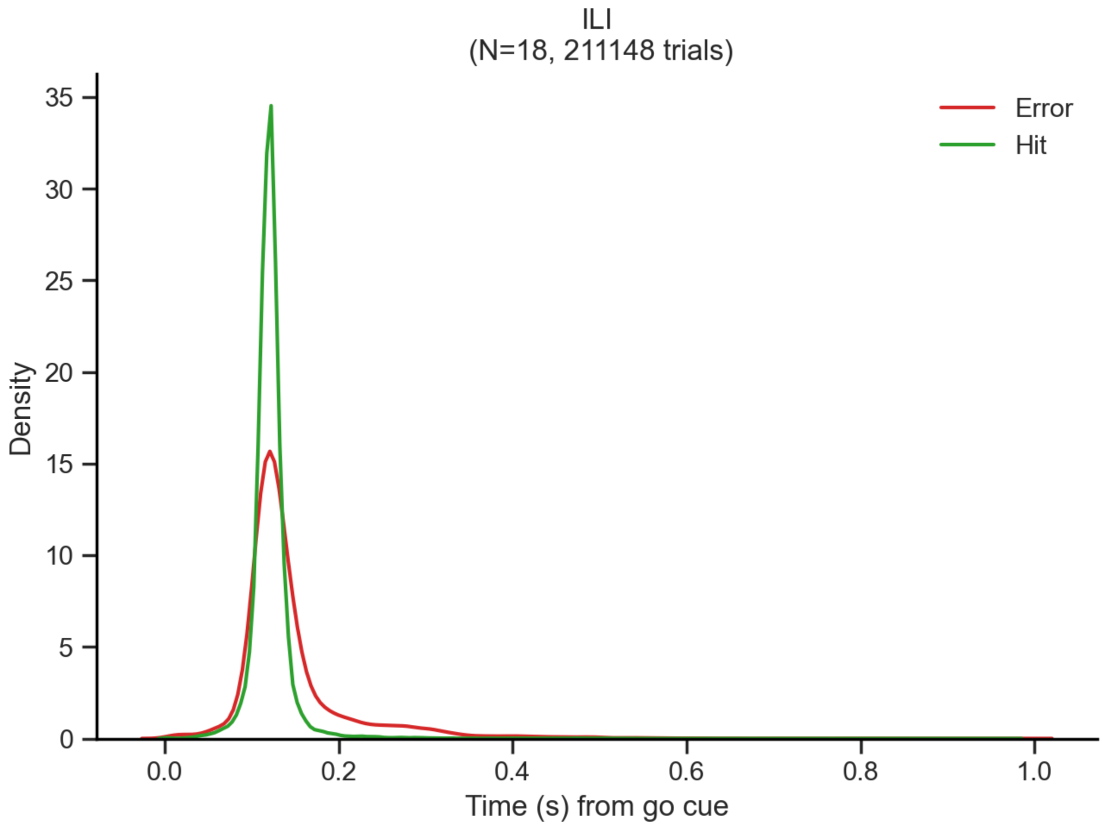
7.0.2 Splits

ILD

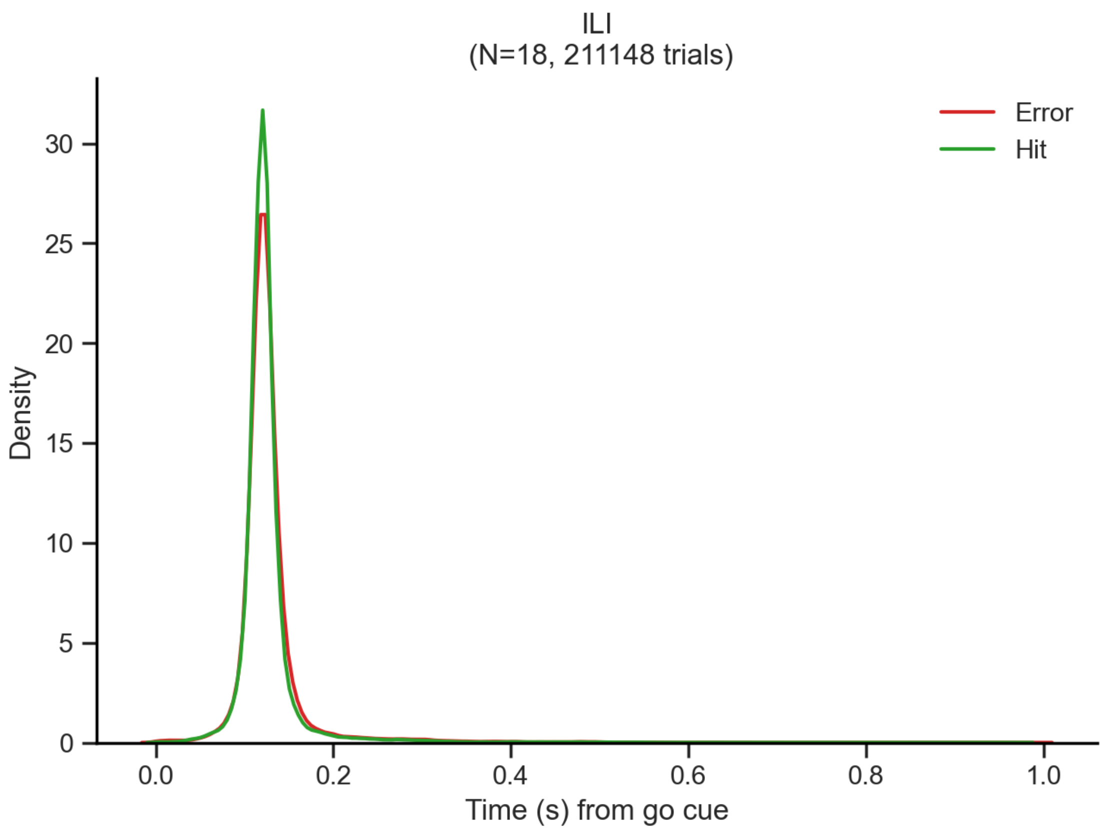
```
ILD 70: peak ILI = 0.122 s
ILD 8: peak ILI = 0.122 s
ILD 4: peak ILI = 0.120 s
ILD 2: peak ILI = 0.121 s
ILD 0: peak ILI = 0.119 s
Mean ILI = 0.121 s
```


ILI distribution
|ILD|
70
8
4
2
0
Density
ILI
0
5
10
15
20
25
30
35
0.0
0.2
0.4
0.6
0.8
1.0

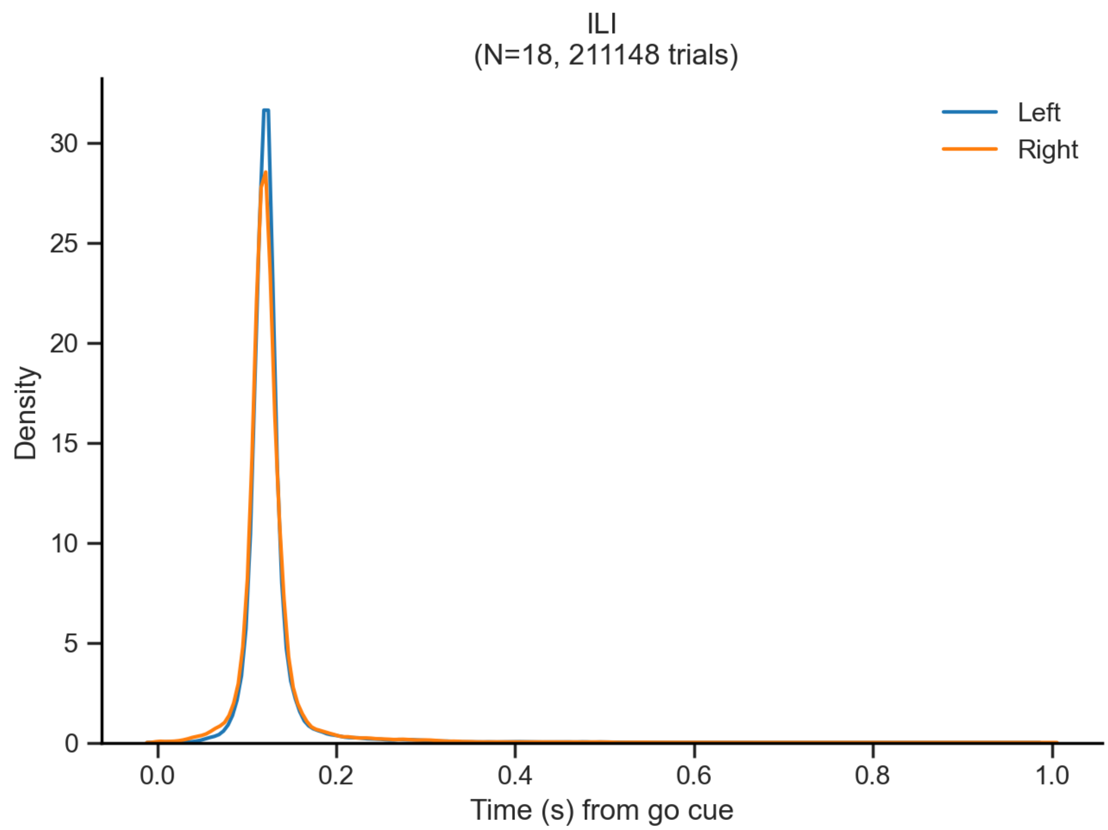
Outcome



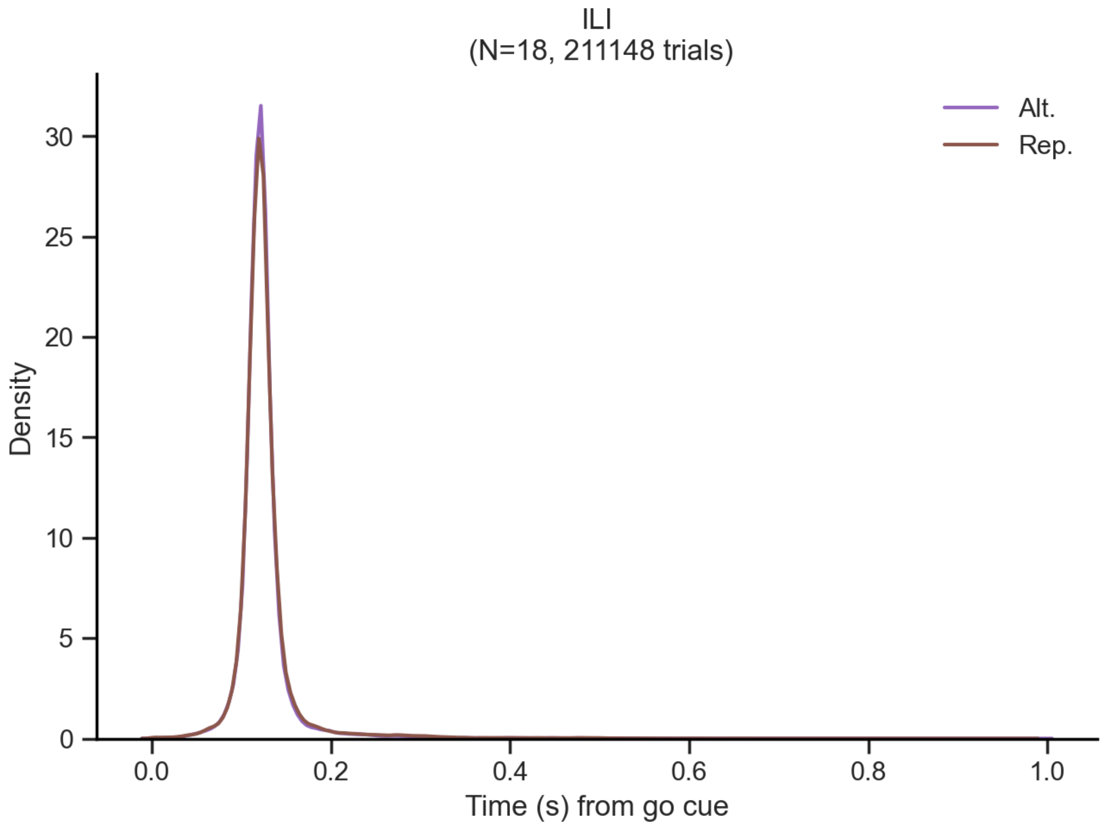
Previous Outcome


ILI
(N=18, 211148 trials)
Error
Hit
Density
30
25
20
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
Time (s) from go cue

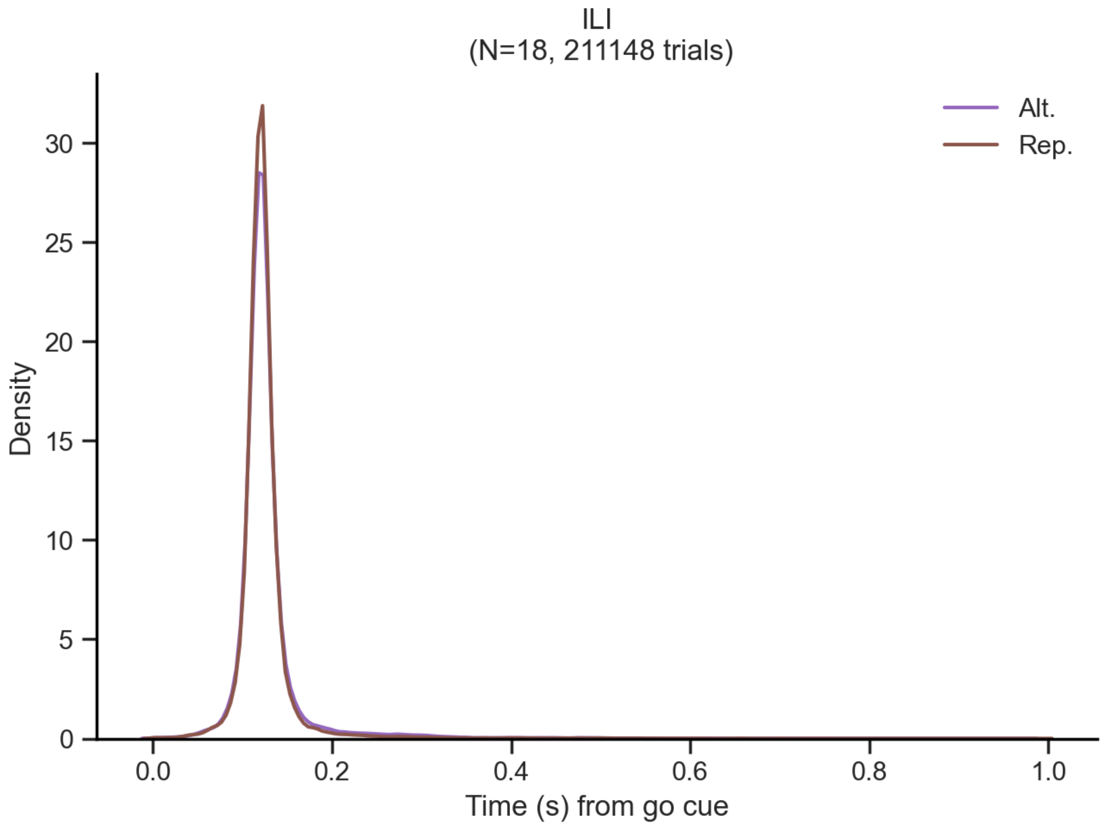
Choice



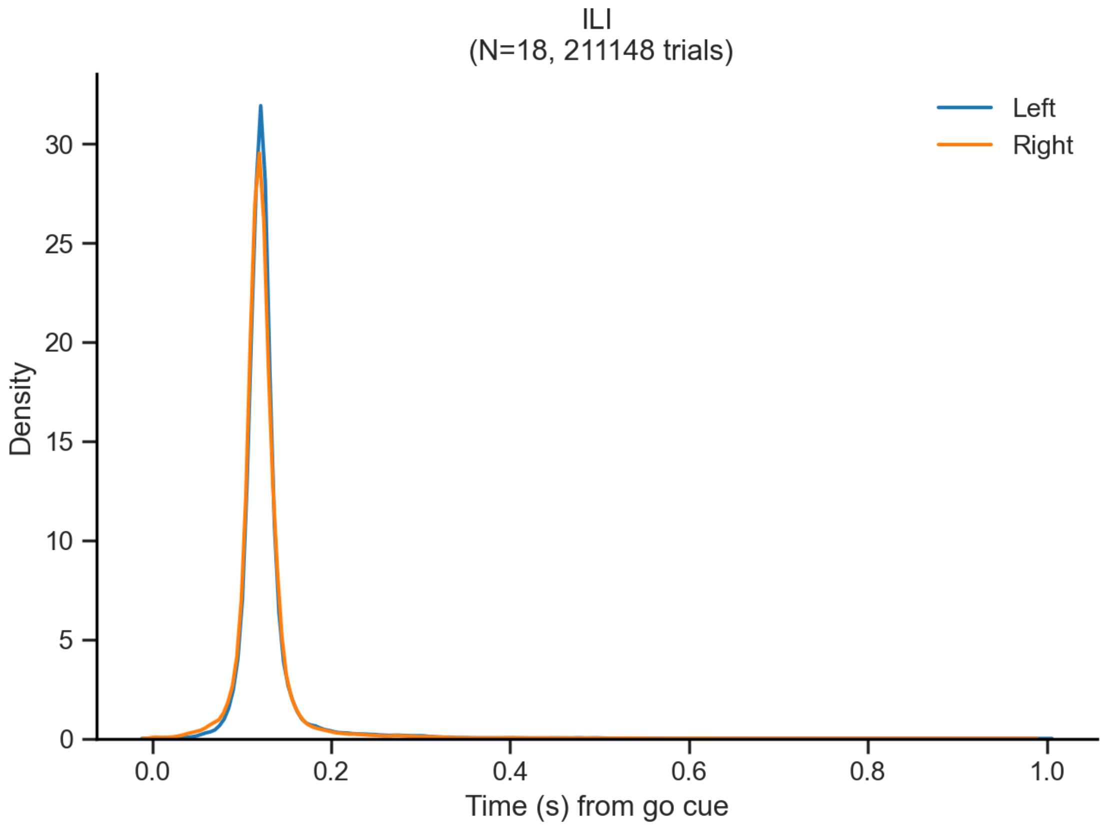
Repeat choice



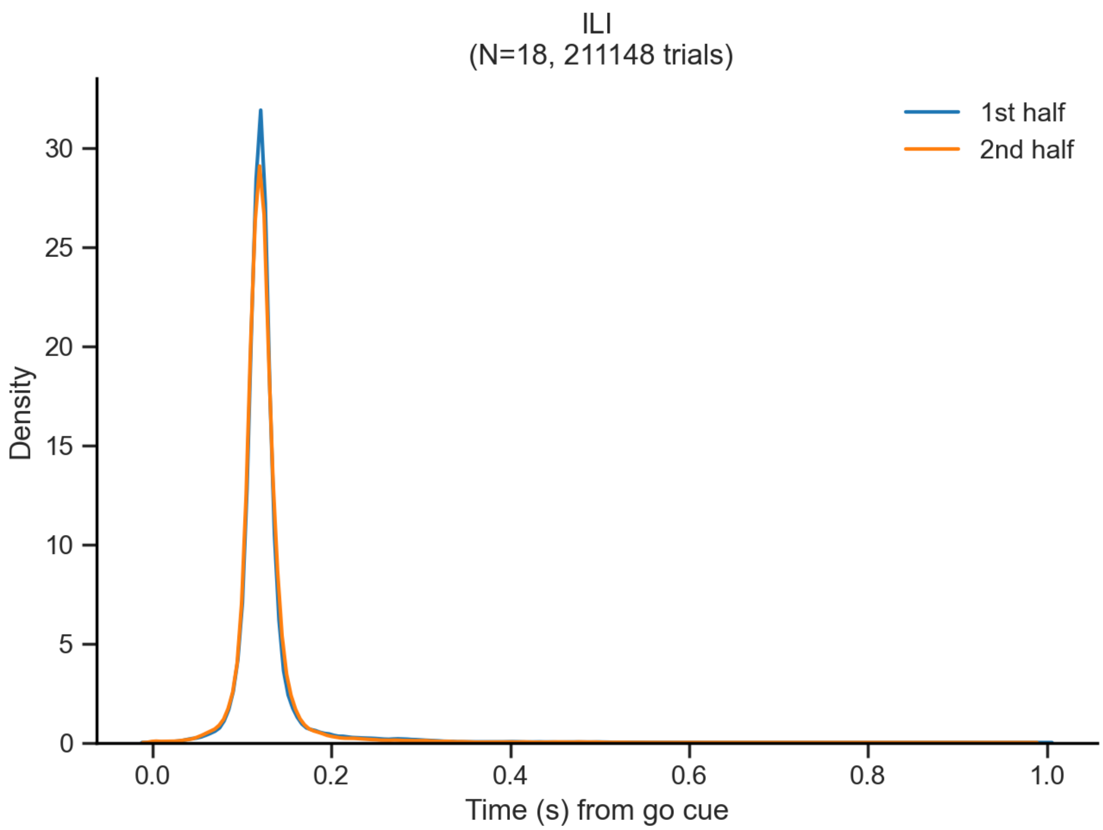
Repeat trial


ILI
(N=18, 211148 trials)
Alt.
Rep.
Density
30
25
20
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
Time (s) from go cue

Stimulus



1st vs 2nd session half



8 Save all notebook's figures

```
[NbConvertApp] Converting notebook notebook.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 61 image(s).
[NbConvertApp] Writing 5559798 bytes to notebook.html
[NbConvertApp] Converting notebook notebook.ipynb to pdf
[NbConvertApp] Support files will be in notebook_files\
[NbConvertApp] Making directory .\notebook_files
[NbConvertApp] Writing 45823 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | b had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
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---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[51], line 8
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```
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 ↪--no-input --no-prompt')
      7 # Export all figures
----> 8 save_notebook_outputs("notebook.ipynb")

NameError: name 'save_notebook_outputs' is not defined
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